

Combinatorial Mathematics



Chapter 8

Advanced Counting Techniques

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-  8.1 Recurrence Relations
 -  8.2 Solving Linear Recurrence Relations
 -  8.3 Divide-and-Conquer Algorithms and Recurrence Relations
 -  8.4 Generating Functions
 -  8.5 Inclusion—Exclusion
 -  8.6 Applications of Inclusion—Exclusion



Recurrence Relations

(recalling definitions from Chapter 2)

Definition: A *recurrence relation* for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the previous terms of the sequence, namely, $a_0, a_1, \dots, a_{n-1}$, for all integers n with $n \geq n_0$, where n_0 is a nonnegative integer.

- A sequence is called a *solution* of a recurrence relation if its terms satisfy the recurrence relation.
- The *initial conditions* for a sequence specify the terms that precede the first term where the recurrence relation takes effect.



8.1 Recurrence Relations(遞迴關係)

- **Example 1.** Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} - a_{n-2}$ for $n=2,3,\dots$, and suppose that $a_0=3$, and $a_1=5$.

Here $a_0=3$ and $a_1=5$ are the initial conditions.

By the recurrence relation,

$$a_2 = a_1 - a_0 = 2$$

$$a_3 = a_2 - a_1 = -3$$

$$a_4 = a_3 - a_2 = -5$$

:

Q1: Applications ?

Q2: Are there better ways for computing the terms of $\{a_n\}$?

- 
- **Example:** Suppose that the *number* of bacteria in a colony *doubles every hour*. If a colony *begins with five bacteria*, how many will be present in n hours?

- Let a_n be the number of bacteria at the end of n hours.

Because the number of bacteria doubles every hour, the relationship $a_n = 2a_{n-1}$ holds whenever n is a positive integer.

This recurrence relation, together with the initial condition $a_0 = 5$, uniquely determines a_n for all nonnegative integers n .

We can find a formula for a_n using the iterative approach followed in Chapter 2, namely that

$$a_n = 5 \cdot 2^n \text{ for all nonnegative integers } n.$$

✂ Modeling with Recurrence Relations

We can use recurrence relations to model (describe) a wide variety of problems.

Example 3. Compound Interest (複利)

Suppose that a person deposits(存款) \$10000 in a saving account at a bank yielding 11% per year with interest compounded annually.

How much will be in the account after 30 years ?

Sol : Let P_n denote the amount in the account after n years.

$$\begin{aligned} P_n &= P_{n-1} + 0.11 \times P_{n-1} = 1.11 \times P_{n-1}, & P_0 &= 10000 \\ \therefore P_{30} &= 1.11 \times P_{29} = (1.11)^2 \times P_{28} = \dots = (1.11)^{30} \times P_0 \\ &= 228922.97 \end{aligned}$$



Rabbits and the Fibonacci Numbers

Example: A young pair of rabbits (one of each gender) is placed on an island. A pair of rabbits does not breed until they are 2 months old. *After they are 2 months old, each pair of rabbits produces another pair each month.* Find a recurrence relation for the number of pairs of rabbits on the island after n months, assuming that *rabbits never die.*

*This is the original problem considered by **Leonardo Pisano (Fibonacci)** in the thirteenth century.*

Rabbits and the Fibonacci Numbers

Reproducing pairs (at least two months old)	Young pairs (less than two months old)	Month	f_{n-2}	f_{n-1}	f_n
			Reproducing pairs	Young pairs	Total pairs
		1	0	1	1
		2	0	1	1
		3	1	1	2
		4	1	2	3
		5	2	3	5
		6	3	5	8
					

Modeling the Population Growth of Rabbits on an Island

[Jump to long description](#)



Rabbits and the Fibonacci Numbers

Solution: Let f_n be the number of pairs of rabbits after n months.

- There are $f_1 = 1$ pairs of rabbits on the island at the end of the first month.
- We also have $f_2 = 1$ because the pair does not breed during the first month.
- To find the number of pairs on the island after n months, add the number on the island after the previous month, f_{n-1} , and the number of newborn pairs, which equals f_{n-2} , because each newborn pair comes from a pair at least two months old.

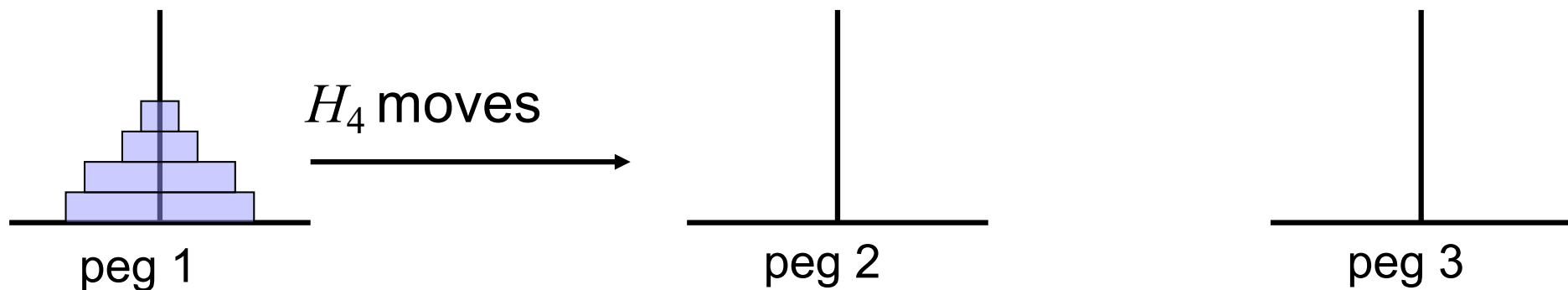
Consequently the sequence $\{f_n\}$ satisfies the recurrence relation $f_n = f_{n-1} + f_{n-2}$ for $n \geq 3$ with the initial conditions $f_1 = 1$ and $f_2 = 1$.

The number of pairs of rabbits on the island after n months is given by the n_{th} Fibonacci number.

Example 5. (The Tower of Hanoi by the French mathematician Édouard Lucas)

The rules of the puzzle allow disks to be moved one at a time from one peg to another as long as **a disk is never placed on top of a smaller disk**. Let H_n denote the number of moves needed to solve the Tower of Hanoi problem with n disks. Set up a recurrence relation for the sequence $\{H_n\}$.

目標： n 個 disk 都從 peg 1 移到 peg 2



Sol : $H_n = 2H_{n-1} + 1, H_1 = 1$

($n-1$ 個 disk 先從 peg 1 $\rightarrow$ peg 3, 第 n 個 disk 從 peg 1 $\rightarrow$ peg 2, $n-1$ 個 disk 再從 peg 3 $\rightarrow$ peg 2)

上例中 $H_n = 2H_{n-1} + 1$, $H_1 = 1$

$$\therefore H_n = 2H_{n-1} + 1$$

$$= 2(2H_{n-2} + 1) + 1$$

$$= 2^2 H_{n-2} + 2 + 1$$

$$= 2^2 (2H_{n-3} + 1) + 2 + 1$$

$$= 2^3 H_{n-3} + (2^2 + 2 + 1)$$

⋮

$$= 2^{n-1} H_1 + (2^{n-2} + 2^{n-3} + \dots + 1)$$

$$= 2^{n-1} + 2^{n-2} + \dots + 1$$

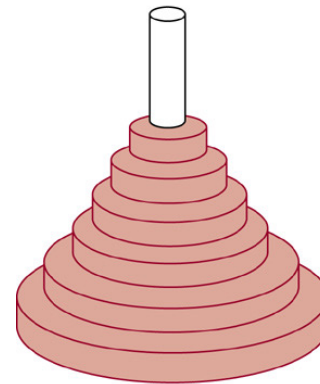
$$= \frac{2^n - 1}{2 - 1} = 2^n - 1$$

A myth created to accompany the puzzle tells of a tower in Hanoi where monks are transferring 64 gold disks from one peg to another, according to the rules of the puzzle. The myth says that the world will end when they finish the puzzle. How long after the monks started will the world end if the monks take one second to move a disk?

From the explicit formula, the monks require

$$2^{64} - 1 = 18,446,744,073,709,551,615$$

moves to transfer the disks. Making one move per second, it will take them more than 500 billion years to complete the transfer, so the world should survive a while longer than it already has. ▶



Peg 1

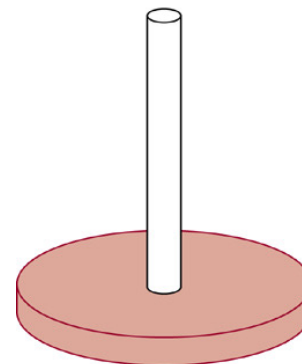


Peg 2



Peg 3

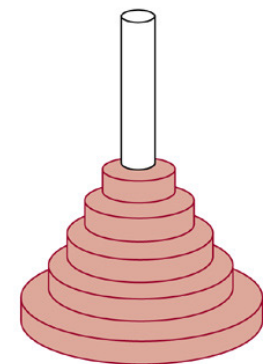
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Peg 1



Peg 2



Peg 3

Example 6. Find a recurrence relation and give initial conditions for the number of **bit strings of length n** that *do not have two consecutive 0s*.

How many such bit strings are there of length 5 ?

Sol :

Let a_n be the number of bit strings of length n that do not have two consecutive 0s.

1
2
...
 $n-3$
 $n-2$
 $n-1$
 n

$$\begin{array}{lcl}
 a_n \left\{ \begin{array}{ll}
 \overline{a_{n-1} \text{ 種}} & \mathbf{1} \\
 \overline{a_{n-2} \text{ 種}} & \mathbf{1 \quad 0}
 \end{array} \right. & \begin{array}{l}
 \therefore a_n = a_{n-1} + a_{n-2}, \quad n \geq 3 \\
 a_1 = 2 \text{ (string : 0,1)} \\
 a_2 = 3 \text{ (string : 01,10,11)}
 \end{array}
 \end{array}$$

$$\therefore a_3 = a_2 + a_1 = 5, \quad a_4 = 8, \quad a_5 = 13$$

Example 7. (Codeword enumeration)

A computer system considers a **string of decimal digits** a **valid codeword** if it contains *an even number of 0 digits*. Let a_n be the number of valid n -digit codewords.

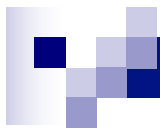
Find a recurrence relation for a_n .

Sol :

1	2	3	...	n-1	n	$\therefore a_n = 9a_{n-1} + (10^{n-1} - a_{n-1})$ $= 8a_{n-1} + 10^{n-1}, n \geq 2$ $a_1 = 9$
					1~9	
a_{n-1} 種						
$10^{n-1} - a_{n-1}$ 種					0	

a_n

$\left\{ \begin{array}{l} \\ \\ \\ \end{array} \right.$



Solving Linear Recurrence Relations

Section 8.2



Linear Homogeneous Recurrence Relations

Definition: A *linear homogeneous recurrence relation of degree k with constant coefficients* is a recurrence relation of the form $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$, where $c_1, c_2, \dots, c_k$ are real numbers, and $c_k \neq 0$

- it is **linear** because the right-hand side is a sum of the previous terms of the sequence each multiplied by a function of n .
- it is **homogeneous** because no terms occur that are not multiples of the a_j s. Each coefficient is a constant.
- the **degree** is k because a_n is expressed in terms of the previous k terms of the sequence.

By strong induction, a sequence satisfying such a recurrence relation is uniquely determined by the recurrence relation and the k initial conditions $a_0 = C_0, a_1 = C_1, \dots, a_{k-1} = C_{k-1}$.



Examples of Linear Homogeneous Recurrence Relations

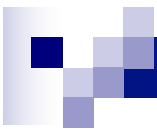
$P_n = (1.11) P_{n-1}$ linear homogeneous recurrence relation of degree one

$f_n = f_{n-1} + f_{n-2}$ linear homogeneous recurrence relation of degree two

$a_n = a_{n-1} + a_{n-2}^2$ the square for a_{n-2} means ***not linear***

$H_n = 2H_{n-1} + 1$ 1 is ***not homogeneous***

$B_n = nB_{n-1}$ coefficient n is ***not constants***



Solving Linear Homogeneous Recurrence Relations

The basic approach is to look for solutions of the form $a_n = r^n$, where r is a constant.

Note that $a_n = r^n$ is a solution to the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} \text{ if and only if}$$

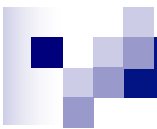
$$r^n = c_1 r^{n-1} + c_2 r^{n-2} + \dots + c_k r^{n-k}$$

Algebraic manipulation yields the ***characteristic equation***:

$$r^k - c_1 r^{k-1} - c_2 r^{k-2} - \dots - c_{k-1} r - c_k = 0$$

The sequence $\{a_n\}$ with $a_n = r^n$ is a solution if and only if r is a solution to the characteristic equation.

The solutions to the characteristic equation are called the ***characteristic roots*** of the recurrence relation. The roots are used to give an **explicit formula** for all the solutions of the recurrence relation.



Solving Linear Homogeneous Recurrence Relations of Degree Two

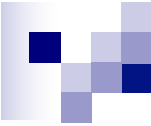
Theorem 1.

Let $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ be a recurrence relation with $c_1, c_2 \in \mathbf{R}$.

If $r^2 - c_1 r - c_2 = 0$ (稱為characteristic equation) has **two distinct roots** r_1 and r_2 .

Then the solution of a_n is $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$, for $n=0,1,2,\dots$, where α_1, α_2 are constants.

(α_1, α_2 可利用 a_0, a_1 算出)



Proof: We must do two things to prove the theorem. First, it must be shown that if r_1 and r_2 are the roots of the characteristic equation, and α_1 and α_2 are constants, then the sequence $\{a_n\}$ with $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$ is a solution of the recurrence relation. Second, it must be shown that if the sequence $\{a_n\}$ is a solution, then $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$ for some constants α_1 and α_2 .

Now we will show that if $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$, then the sequence $\{a_n\}$ is a solution of the recurrence relation. Because r_1 and r_2 are roots of $r^2 - c_1 r - c_2 = 0$, it follows that $r_1^2 = c_1 r_1 + c_2$, $r_2^2 = c_1 r_2 + c_2$.

From these equations, we see that

$$\begin{aligned} c_1 a_{n-1} + c_2 a_{n-2} &= c_1 (\alpha_1 r_1^{n-1} + \alpha_2 r_2^{n-1}) + c_2 (\alpha_1 r_1^{n-2} + \alpha_2 r_2^{n-2}) \\ &= \alpha_1 r_1^{n-2} (c_1 r_1 + c_2) + \alpha_2 r_2^{n-2} (c_1 r_2 + c_2) \\ &= \alpha_1 r_1^{n-2} r_1^2 + \alpha_2 r_2^{n-2} r_2^2 \\ &= \alpha_1 r_1^n + \alpha_2 r_2^n \\ &= a_n. \end{aligned}$$

This shows that the sequence $\{a_n\}$ with $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$ is a solution of the recurrence relation.

To show that every solution $\{a_n\}$ of the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ has $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$ for $n = 0, 1, 2, \dots$, for some constants α_1 and α_2 , suppose that $\{a_n\}$ is a solution of the recurrence relation, and the initial conditions $a_0 = C_0$ and $a_1 = C_1$ hold. It will be shown that there are constants α_1 and α_2 such that the sequence $\{a_n\}$ with $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$ satisfies these same initial conditions. This requires that

$$\begin{aligned} a_0 &= C_0 = \alpha_1 + \alpha_2, \\ a_1 &= C_1 = \alpha_1 r_1 + \alpha_2 r_2. \end{aligned}$$

We can solve these two equations for α_1 and α_2 . From the first equation it follows that $\alpha_2 = C_0 - \alpha_1$. Inserting this expression into the second equation gives

$$C_1 = \alpha_1 r_1 + (C_0 - \alpha_1) r_2.$$

Hence,

$$C_1 = \alpha_1 (r_1 - r_2) + C_0 r_2.$$

This shows that

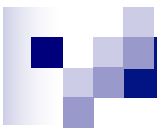
$$\alpha_1 = \frac{C_1 - C_0 r_2}{r_1 - r_2}$$

and

$$\alpha_2 = C_0 - \alpha_1 = C_0 - \frac{C_1 - C_0 r_2}{r_1 - r_2} = \frac{C_0 r_1 - C_1}{r_1 - r_2},$$

where these expressions for α_1 and α_2 depend on the fact that $r_1 \neq r_2$. (When $r_1 = r_2$, this theorem is not true.) Hence, with these values for α_1 and α_2 , the sequence $\{a_n\}$ with $\alpha_1 r_1^n + \alpha_2 r_2^n$ satisfies the two initial conditions.

We know that $\{a_n\}$ and $\{\alpha_1 r_1^n + \alpha_2 r_2^n\}$ are both solutions of the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ and both satisfy the initial conditions when $n = 0$ and $n = 1$. Because there is a unique solution of a linear homogeneous recurrence relation of degree two with two initial conditions, it follows that the two solutions are the same, that is, $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$ for all nonnegative integers n . We have completed the proof by showing that a solution of the linear homogeneous recurrence relation with constant coefficients of degree two must be of the form $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$, where α_1 and α_2 are constants. ◀



Example 3.

What's the solution of the recurrence relation

$$a_n = a_{n-1} + 2a_{n-2}$$

with $a_0=2$ and $a_1=7$?

Sol :

The characteristic equation is $r^2 - r - 2=0$.

Its two roots are $r_1=2$ and $r_2=-1$.

Hence $a_n = \alpha_1 \times 2^n + \alpha_2 \times (-1)^n$.

$$\because a_0 = \alpha_1 + \alpha_2 = 2, \quad a_1 = 2\alpha_1 - \alpha_2 = 7$$

$$\because \alpha_1 = 3, \quad \alpha_2 = -1$$

$$\Rightarrow a_n = 3 \times 2^n - (-1)^n.$$

$$\text{驗算 : } a_2 = a_1 + 2a_0 = 11$$

$$a_2 = 3 \times 2^2 - 1 = 11$$

Example 4. Find an explicit formula for the Fibonacci numbers.

Sol :

$$f_n = f_{n-1} + f_{n-2}, n \geq 2, f_0 = 0, f_1 = 1.$$

The characteristic equation is $r^2 - r - 1 = 0$.

Its two roots are $r_1 = \frac{1 + \sqrt{5}}{2}$, $r_2 = \frac{1 - \sqrt{5}}{2}$.

So we have

$$f_n = \alpha_1 \cdot \left(\frac{1 + \sqrt{5}}{2}\right)^n + \alpha_2 \left(\frac{1 - \sqrt{5}}{2}\right)^n$$

$$\because f_0 = \alpha_1 + \alpha_2 = 0, f_1 = \alpha_1 \left(\frac{1 + \sqrt{5}}{2}\right) + \alpha_2 \left(\frac{1 - \sqrt{5}}{2}\right) = 1$$

$$\therefore \alpha_1 = \frac{1}{\sqrt{5}}, \alpha_2 = \frac{-1}{\sqrt{5}}$$

$$\Rightarrow f_n = \frac{1}{\sqrt{5}} \cdot \left(\frac{1 + \sqrt{5}}{2}\right)^n + \frac{-1}{\sqrt{5}} \left(\frac{1 - \sqrt{5}}{2}\right)^n$$

$$ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



An Explicit Formula for the Fibonacci Numbers

We can use Theorem 1 to find an explicit formula for the *Fibonacci numbers*. The sequence of Fibonacci numbers satisfies the recurrence relation $f_n = f_{n-1} + f_{n-2}$ with the initial conditions: $f_0 = 0$ and $f_1 = 1$.

Solution: The roots of the characteristic equation $r^2 - r - 1 = 0$ are

$$r_1 = \frac{1 + \sqrt{5}}{2}$$

$$r_2 = \frac{1 - \sqrt{5}}{2}$$



Fibonacci Numbers

Therefore by Theorem 1

$$f_n = \alpha_1 \left(\frac{1+\sqrt{5}}{2} \right)^n + \alpha_2 \left(\frac{1-\sqrt{5}}{2} \right)^n$$

for some constants α_1 and α_2 .

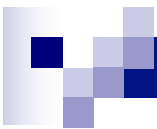
Using the initial conditions $f_0 = 0$ and $f_1 = 1$, we have

$$\begin{aligned} f_0 &= \alpha_1 + \alpha_2 = 0 \\ f_1 &= \alpha_1 \left(\frac{1+\sqrt{5}}{2} \right) + \alpha_2 \left(\frac{1-\sqrt{5}}{2} \right) = 1. \end{aligned}$$

Solving, we obtain $\alpha_1 = \frac{1}{\sqrt{5}}, \quad \alpha_2 = -\frac{1}{\sqrt{5}}.$

Hence,

$$f_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n.$$



The Solution when there is a Repeated Root

Thm 2.

Let $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ be a recurrence relation with $c_1, c_2 \in \mathbf{R}$.

If $r^2 - c_1 r - c_2 = 0$ has only one root r_0 .

Then the solution of a_n is

$$a_n = \alpha_1 \cdot r_0^n + \alpha_2 \cdot n \cdot r_0^n$$

for $n=0,1,2,\dots$, where α_1 and α_2 are constants.



Example 5.

What's the solution of $a_n = 6a_{n-1} - 9a_{n-2}$
with $a_0=1$ and $a_1=6$?

Sol :

The root of $r^2 - 6r + 9 = 0$ is $r_0 = 3$.

Hence $a_n = \alpha_1 \cdot 3^n + \alpha_2 \cdot n \cdot 3^n$.

$$\because a_0 = \alpha_1 = 1$$

$$a_1 = 3\alpha_1 + 3\alpha_2 = 6$$

$$\therefore \alpha_1 = 1 \text{ and } \alpha_2 = 1$$

$$\Rightarrow a_n = 3^n + n \cdot 3^n$$

驗算 : $a_2 = 6a_1 - 9a_0 = 27$ $a_2 = 3^2 + 2 \times 3^2 = 27$
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Solving Linear Homogeneous Recurrence Relations of Arbitrary Degree

Thm 3.

Let $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$ be a recurrence relation with $c_1, c_2, \dots, c_k \in \mathbf{R}$.

If $r^k - c_1 r^{k-1} - c_2 r^{k-2} - \dots - c_k = 0$ has k **distinct roots** $r_1, r_2, \dots, r_k$.

Then the solution of a_n is

$$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n + \dots + \alpha_k r_k^n, \text{ for } n = 0, 1, 2, \dots$$

where $\alpha_1, \alpha_2, \dots, \alpha_k$ are constants.

Example 6 ($k = 3$)

Find the solution of $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ with initial conditions $a_0=2$, $a_1=5$ and $a_2=15$.

Sol :

The roots of $r^3 - 6r^2 + 11r - 6 = 0$ are

$$r_1 = 1, r_2 = 2, \text{ and } r_3 = 3$$

$$\therefore a_n = \alpha_1 \cdot 1^n + \alpha_2 \cdot 2^n + \alpha_3 \cdot 3^n$$

$$\begin{array}{l} \therefore a_0 = \alpha_1 + \alpha_2 + \alpha_3 = 2 \\ a_1 = \alpha_1 + 2\alpha_2 + 3\alpha_3 = 5 \\ a_2 = \alpha_1 + 4\alpha_2 + 9\alpha_3 = 15 \end{array} \left. \vphantom{\begin{array}{l} a_0 \\ a_1 \\ a_2 \end{array}} \right\} \begin{array}{l} \alpha_1 = 1, \\ \alpha_2 = -1, \\ \alpha_3 = 2 \end{array}$$

$$\therefore a_n = 1 - 2^n + 2 \cdot 3^n$$

$$\begin{array}{l} \text{驗算 : } a_3 = 6a_2 - 11a_1 + 6a_0 = 47 \\ a_3 = 1 - 2^3 + 2 \cdot 3^3 = 47 \end{array}$$



The General Case with Repeated Roots Allowed

Thm 4.

Let $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$ be a recurrence relation with $c_1, c_2, \dots, c_k \in \mathbf{R}$.

If $r^k - c_1 r^{k-1} - c_2 r^{k-2} - \dots - c^k = 0$

has t **distinct** roots $r_1, r_2, \dots, r_t$ with **multiplicities** $m_1, m_2, \dots, m_t$ respectively, where $m_i \geq 1, \forall i$, and $m_1 + m_2 + \dots + m_t = k$,

then

(接下一頁)



$$\left\{ \begin{array}{l}
 a_n = (\alpha_{1,0} + \alpha_{1,1} \cdot n + \dots + \alpha_{1,m_1-1} \cdot n^{m_1-1}) r_1^n \\
 + \dots \\
 + (\alpha_{i,0} + \alpha_{i,1} \cdot n + \dots + \alpha_{i,m_i-1} \cdot n^{m_i-1}) \cdot r_i^n \\
 + \dots \\
 + (\alpha_{t,0} + \alpha_{t,1} \cdot n + \dots + \alpha_{t,m_t-1} \cdot n^{m_t-1}) \cdot r_t^n
 \end{array} \right.$$

where $\alpha_{i,j}$ are constants.

$$(1 \leq i \leq t, 0 \leq j \leq m_i-1)$$



補充說明：

若特徵方程式的root為：

- 1 (2重根),
- 2 (3重根),
- 3 (無重根)

則通解為：

$$a_n = (\alpha_1 + \alpha_2 \cdot n) \cdot 1^n + (\alpha_3 + \alpha_4 \cdot n + \alpha_5 \cdot n^2) \cdot (-2)^n + \alpha_6 \cdot 3^n$$

(變數 α 的下標從1開始排起，只要不重複就好)



EXAMPLE 7 Suppose that the roots of the characteristic equation of a linear homogeneous recurrence relation are 2, 2, 2, 5, 5, and 9 (that is, there are three roots, the root 2 with multiplicity three, the root 5 with multiplicity two, and the root 9 with multiplicity one). What is the form of the general solution?

Solution: By Theorem 4, the general form of the solution is

$$(\alpha_{1,0} + \alpha_{1,1}n + \alpha_{1,2}n^2)2^n + (\alpha_{2,0} + \alpha_{2,1}n)5^n + \alpha_{3,0}9^n.$$




Example 8. Find the solution to the recurrence relation $a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}$ with initial conditions $a_0 = 1$, $a_1 = -2$ and $a_2 = -1$.

Sol :

$r^3 + 3r^2 + 3r + 1 = 0$ has a single root $r_0 = -1$ of multiplicity three.

$$\therefore a_n = (\alpha_1 + \alpha_2 n + \alpha_3 n^2) r_0^n = (\alpha_1 + \alpha_2 n + \alpha_3 n^2)(-1)^n$$

$$\therefore a_0 = \alpha_1 = 1$$

$$a_1 = (\alpha_1 + \alpha_2 + \alpha_3) \cdot (-1) = -2$$

$$a_2 = (\alpha_1 + \alpha_2 + \alpha_3) = -1$$

$$\therefore \alpha_1 = 1, \alpha_2 = 3, \alpha_3 = -2$$

$$\Rightarrow a_n = (1 + 3n - 2n^2) \cdot (-1)^n$$

$$\begin{aligned} \text{驗算 : } a_3 &= -3a_2 - 3a_1 - a_0 = 8 \\ a_3 &= (1 + 3 \cdot 3 - 2 \cdot 3^2) \cdot (-1)^3 = 8 \end{aligned}$$

Linear Nonhomogeneous Recurrence Relations with Constant Coefficients

The following are linear nonhomogeneous recurrence relations with constant coefficients:

$$a_n = a_{n-1} + 2^n,$$

$$a_n = a_{n-1} + a_{n-2} + n^2 + n + 1,$$

$$a_n = 3a_{n-1} + n3^n,$$

$$a_n = a_{n-1} + a_{n-2} + a_{n-3} + n!$$

where the following are the associated linear homogeneous recurrence relations, respectively:

$$a_n = a_{n-1},$$

$$a_n = a_{n-1} + a_{n-2},$$

$$a_n = 3a_{n-1},$$

$$a_n = a_{n-1} + a_{n-2} + a_{n-3}$$



Solving Linear Nonhomogeneous Recurrence Relations with Constant Coefficients

Theorem 5: If $\{a_n^{(p)}\}$ is a particular solution of the nonhomogeneous linear recurrence relation with constant coefficients

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} + F(n),$$

then every solution is of the form $\{a_n^{(p)} + a_n^{(h)}\}$, where $\{a_n^{(h)}\}$ is a solution of the associated homogeneous recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$$



EXAMPLE 10 Find all solutions of the recurrence relation $a_n = 3a_{n-1} + 2n$. What is the solution with $a_1 = 3$?

Solution: To solve this linear nonhomogeneous recurrence relation with constant coefficients, we need to solve its associated linear homogeneous equation and to find a particular solution for the given nonhomogeneous equation. The associated linear homogeneous equation is $a_n = 3a_{n-1}$. Its solutions are $a_n^{(h)} = \alpha 3^n$, where α is a constant.

We now find a particular solution. Because $F(n) = 2n$ is a polynomial in n of degree one, a reasonable trial solution is a linear function in n , say, $p_n = cn + d$, where c and d are constants. To determine whether there are any solutions of this form, suppose that $p_n = cn + d$ is such a solution. Then the equation $a_n = 3a_{n-1} + 2n$ becomes $cn + d = 3(c(n-1) + d) + 2n$. Simplifying and combining like terms gives $(2 + 2c)n + (2d - 3c) = 0$. It follows that $cn + d$ is a solution if and only if $2 + 2c = 0$ and $2d - 3c = 0$. This shows that $cn + d$ is a solution if and only if $c = -1$ and $d = -3/2$. Consequently, $a_n^{(p)} = -n - 3/2$ is a particular solution.

By Theorem 5 all solutions are of the form

$$a_n = a_n^{(p)} + a_n^{(h)} = -n - \frac{3}{2} + \alpha \cdot 3^n,$$

where α is a constant.

To find the solution with $a_1 = 3$, let $n = 1$ in the formula we obtained for the general solution. We find that $3 = -1 - 3/2 + 3\alpha$, which implies that $\alpha = 11/6$. The solution we seek is $a_n = -n - 3/2 + (11/6)3^n$. 



8.3 Divide-and-Conquer Algorithms and Recurrence Relations

$$f(n) = af(n/b) + g(n).$$



EXAMPLE 1 **Binary Search** We introduced a binary search algorithm in Section 3.1. This binary search algorithm reduces the search for an element in a search sequence of size n to the binary search for this element in a search sequence of size $n/2$, when n is even. (Hence, the problem of size n has been reduced to *one* problem of size $n/2$.) Two comparisons are needed to implement this reduction (one to determine which half of the list to use and the other to determine whether any terms of the list remain). Hence, if $f(n)$ is the number of comparisons required to search for an element in a search sequence of size n , then



$$f(n) = f(n/2) + 2$$

EXAMPLE 2 **Finding the Maximum and Minimum of a Sequence** Consider the following algorithm for locating the maximum and minimum elements of a sequence $a_1, a_2, \dots, a_n$. If $n = 1$, then a_1 is the maximum and the minimum. If $n > 1$, split the sequence into two sequences, either where both have the same number of elements or where one of the sequences has one more element than the other. The problem is reduced to finding the maximum and minimum of each of the two smaller sequences. The solution to the original problem results from the comparison of the separate maxima and minima of the two smaller sequences to obtain the overall maximum and minimum.

Let $f(n)$ be the total number of comparisons needed to find the maximum and minimum elements of the sequence with n elements. We have shown that a problem of size n can be reduced into two problems of size $n/2$, when n is even, using two comparisons, one to compare the maxima of the two sequences and the other to compare the minima of the two sequences. This gives the recurrence relation

$$f(n) = 2f(n/2) + 2$$



8.4 Generating Functions.

Def 1. The **generating function** for the sequence $\{a_k\}$ is the infinite (formal) power series.

$$\begin{aligned} G(x) &= a_0 + a_1x + \dots + a_kx^k + \dots \\ &= \sum_{k=0}^{\infty} a_k x^k \end{aligned}$$

(若 $\{a_k\}$ 是finite，可視為是infinite，但後面的term都等於0)



Example 1. Find the generating functions for the sequences $\{a_k\}$ with

(1) $a_k = 3$

(2) $a_k = k+1$

(3) $a_k = 2^k$

Sol :

$$(1) \quad G(x) = \sum_{k=0}^{\infty} a_k x^k = \sum_{k=0}^{\infty} 3x^k$$

$$(2) \quad G(x) = \sum_{k=0}^{\infty} a_k x^k = \sum_{k=0}^{\infty} (k+1)x^k$$

$$(3) \quad G(x) = \sum_{k=0}^{\infty} a_k x^k = \sum_{k=0}^{\infty} 2^k x^k$$

Example 2. What is the generating function for the sequence 1,1,1,1,1,1 ?

Sol : $a_0 a_1 a_2 a_3 a_4 a_5$ a_6 及之後都=0

$$G(x) = 1 + x + x^2 + \dots + x^5 \quad (\text{expansion, 展開式})$$

$$= \frac{x^6 - 1}{x - 1} \quad (\text{closed form})$$

THEOREM 1 If a and r are real numbers and $r \neq 0$, then

$$\sum_{j=0}^n ar^j = \begin{cases} \frac{ar^{n+1} - a}{r - 1} & \text{if } r \neq 1 \\ (n+1)a & \text{if } r = 1. \end{cases}$$

Example 3.

Let $m \in \mathbf{Z}^+$ and $a_k = \binom{m}{k}$, for $k = 0, 1, \dots, m$.

What is the generating function for the sequence $a_0, a_1, \dots, a_m$?

Sol :

$$G(x) = a_0 + a_1x + a_2x^2 + \dots + a_mx^m$$

$$= \binom{m}{0} + \binom{m}{1}x + \binom{m}{2}x^2 + \dots + \binom{m}{m}x^m$$

$$= (1+x)^m \quad (\text{by 二項式定理})$$

$$(x+y)^n = \binom{n}{0}x^n + \binom{n}{1}x^{n-1}y + \dots + \binom{n}{n-1}xy^{n-1} + \binom{n}{n}y^n = \sum_{j=0}^n \binom{n}{j}x^{n-j}y^j$$

Useful Facts About Power Series

Example 4.

The function $f(x) = \frac{1}{1-x}$ is the generating function of the sequence $1, 1, 1, \dots$, because

$$\sum_{k=0}^{\infty} x^k = 1 + x + x^2 + \dots = \frac{1}{1-x} \text{ when } |x| < 1.$$

THEOREM 1 If a and r are real numbers and $r \neq 0$, then

Example 5.

The function $f(x) = \frac{1}{1-ax}$ is the generating function of the sequence $1, a, a^2, \dots$, because

$$\sum_{k=0}^{\infty} (ax)^k = 1 + ax + a^2x^2 + \dots = \frac{1}{1-ax} \text{ when } |ax| < 1 \text{ for } a \neq 0.$$

$$\sum_{j=0}^n ar^j = \begin{cases} \frac{ar^{n+1} - a}{r - 1} & \text{if } r \neq 1 \\ (n+1)a & \text{if } r = 1. \end{cases}$$



Theorem 1.

$$\text{Let } f(x) = \sum_{k=0}^{\infty} a_k x^k \text{ and } g(x) = \sum_{k=0}^{\infty} b_k x^k.$$

$$\text{Then } f(x) + g(x) = \sum_{k=0}^{\infty} (a_k + b_k) x^k.$$

$$\begin{aligned} f(x) g(x) &= (a_0 + a_1 x + a_2 x^2 + \dots)(b_0 + b_1 x + b_2 x^2 + \dots) \\ &= (a_0 b_0) + (a_0 b_1 + a_1 b_0) x + (a_0 b_2 + a_1 b_1 + a_2 b_0) x^2 + \dots \\ &= \sum_{k=0}^{\infty} \left(\sum_{j=0}^k a_j b_{k-j} \right) x^k \end{aligned}$$



Example 6.

Let $f(x) = \frac{1}{(1-x)^2}$. Use Example 4 to find the

coefficients $a_0, a_1, a_2, \dots$ in the expansion $f(x) = \sum_{k=0}^{\infty} a_k x^k$.

Sol :

$$\frac{1}{1-x} = 1 + x + x^2 + \dots$$

$$\frac{1}{(1-x)^2} = \sum_{k=0}^{\infty} \left(\sum_{j=0}^k a_j b_{k-j} \right) x^k = \sum_{k=0}^{\infty} \left(\sum_{j=0}^k 1 \right) x^k = \sum_{k=0}^{\infty} (k+1) x^k$$

$$\therefore a_k = k+1$$



Def 2.

Let $u \in \mathbf{R}$ and $k \in \mathbf{Z}^+ \cup \{0\}$. Then the extended

binomial coefficient $\binom{u}{k}$ is defined by

$$\binom{u}{k} = \begin{cases} \frac{1}{k!} u(u-1)(u-2)\dots(u-k+1), & \text{if } k > 0 \\ 1 & \text{if } k = 0 \end{cases}$$



Example 7.

Find $\binom{-2}{3}$ and $\binom{\frac{1}{2}}{3}$

Sol :

$$\binom{-2}{3} = \frac{1}{3!}(-2)(-3)(-4) = -4$$

$$\binom{\frac{1}{2}}{3} = \frac{1}{3!}\left(\frac{1}{2}\right)\left(\frac{-1}{2}\right)\left(\frac{-3}{2}\right) = \frac{1}{16}$$

Example 8

When the top parameter is a negative integer, the extended binomial coefficient can be expressed in terms of an ordinary binomial coefficient.

$$\begin{aligned}\binom{-n}{r} &= \frac{\overbrace{(-n)(-n-1)\dots(-n-r+1)}^{\text{totally } r \text{ } (-1)\text{s}}}{r!} \\&= \frac{(-1)^r \cdot (n)(n+1) \dots (n+r-1)}{r!} \\&= \frac{(-1)^r \cdot [(1)(2) \dots (n-1)] \cdot [(n)(n+1) \dots (n+r-1)]}{[(1)(2) \dots (n-1)] \cdot r!} \\&= \frac{(-1)^r (n+r-1)!}{r!(n-1)!} \\&= (-1)^r \binom{n+r-1}{r}\end{aligned}$$



Thm 2. (The Extended Binomial Theorem)

Let $x \in \mathbf{R}$ with $|x| < 1$ and let $u \in \mathbf{R}$, then

$$(1+x)^u = \sum_{k=0}^{\infty} \binom{u}{k} x^k$$

Example 9.

Find the generating functions for $(1+x)^{-n}$ and $(1-x)^{-n}$ where $n \in \mathbf{Z}^+$

Sol : By the Extended Binomial Theorem,

$$\begin{aligned}(1+x)^{-n} &= \sum_{k=0}^{\infty} \binom{-n}{k} x^k = \sum_{k=0}^{\infty} \frac{1}{k!} (-n)(-n-1) \dots (-n-k+1) x^k \\&= \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} (n)(n+1) \dots (n+k-1) x^k \\&= \sum_{k=0}^{\infty} (-1)^k \binom{n+k-1}{k} x^k\end{aligned}$$

$$\text{Note: } \binom{-n}{k} = (-1)^k \binom{n+k-1}{k}$$

By replacing x by $-x$ we have

$$(1-x)^{-n} = \sum_{k=0}^{\infty} \binom{n+k-1}{k} x^k$$

TABLE

TABLE 1 Useful Generating Functions.

$G(x)$	a_k
$(1+x)^n = \sum_{k=0}^n C(n, k)x^k$ $= 1 + C(n, 1)x + C(n, 2)x^2 + \cdots + x^n$	$C(n, k)$
$(1+ax)^n = \sum_{k=0}^n C(n, k)a^k x^k$ $= 1 + C(n, 1)ax + C(n, 2)a^2x^2 + \cdots + a^n x^n$	$C(n, k)a^k$
$(1+x^r)^n = \sum_{k=0}^n C(n, k)x^{rk}$ $= 1 + C(n, 1)x^r + C(n, 2)x^{2r} + \cdots + x^{rn}$	$C(n, k/r)$ if $r \mid k$; 0 otherwise
$\frac{1-x^{n+1}}{1-x} = \sum_{k=0}^n x^k = 1 + x + x^2 + \cdots + x^n$	1 if $k \leq n$; 0 otherwise
$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k = 1 + x + x^2 + \cdots$	1
$\frac{1}{1-ax} = \sum_{k=0}^{\infty} a^k x^k = 1 + ax + a^2x^2 + \cdots$	a^k
$\frac{1}{1-x^r} = \sum_{k=0}^{\infty} x^{rk} = 1 + x^r + x^{2r} + \cdots$	1 if $r \mid k$; 0 otherwise
$\frac{1}{(1-x)^2} = \sum_{k=0}^{\infty} (k+1)x^k = 1 + 2x + 3x^2 + \cdots$	$k+1$
$\frac{1}{(1-x)^n} = \sum_{k=0}^{\infty} C(n+k-1, k)x^k$ $= 1 + C(n, 1)x + C(n+1, 2)x^2 + \cdots$	$C(n+k-1, k) = C(n+k-1, n-1)$
$\frac{1}{(1+x)^n} = \sum_{k=0}^{\infty} C(n+k-1, k)(-1)^k x^k$ $= 1 - C(n, 1)x + C(n+1, 2)x^2 - \cdots$	$(-1)^k C(n+k-1, k) = (-1)^k C(n+k-1, n-1)$
$\frac{1}{(1-ax)^n} = \sum_{k=0}^{\infty} C(n+k-1, k)a^k x^k$ $= 1 + C(n, 1)ax + C(n+1, 2)a^2x^2 + \cdots$	$C(n+k-1, k)a^k = C(n+k-1, n-1)a^k$
$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$	$1/k!$
$\ln(1+x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} x^k = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$	$(-1)^{k+1}/k$

Counting Problems and Generating Functions

Generating functions can be used to count the number of combinations of various types.

Example 10.

Find the number of solutions of $e_1 + e_2 + e_3 = 17$, where e_1, e_2, e_3 are integers with $2 \leq e_1 \leq 5$, $3 \leq e_2 \leq 6$, and $4 \leq e_3 \leq 7$.

Sol : The number of solutions with the indicated constraints is the coefficient of x^{17} in the expansion of

$$(x^2 + x^3 + x^4 + x^5)(x^3 + x^4 + x^5 + x^6)(x^4 + x^5 + x^6 + x^7)$$

(即相當於找 e_1, e_2, e_3 使 $x^{e_1}x^{e_2}x^{e_3} = x^{17}$)

$\therefore (e_1, e_2, e_3) = (4, 6, 7), (5, 5, 7), (5, 6, 6)$ 共3種



Example 11.

In how many different ways can *eight identical cookies* be distributed *among three distinct children* if *each child receives at least two cookies and no more than four cookies*?

Sol : The number of solutions is the coefficient of x^8 in the expansion of

$$(x^2 + x^3 + x^4)^3$$

$$\therefore (c_1, c_2, c_3) = (2, 2, 4), (2, 3, 3), (2, 4, 2), \\ (3, 2, 3), (3, 3, 2), (4, 2, 2) \\ \text{共6種}$$



EXAMPLE 3.1.3 Find the number of solutions to $x_1 + x_2 + x_3 + x_4 = 17$, where $0 \leq x_1 \leq 2$, $0 \leq x_2 \leq 5$, $0 \leq x_3 \leq 5$, $2 \leq x_4 \leq 6$.

We can of course solve this problem using the inclusion-exclusion formula, but we use generating functions. Consider the function

$$(1 + x + x^2)(1 + x + x^2 + x^3 + x^4 + x^5)(1 + x + x^2 + x^3 + x^4 + x^5)(x^2 + x^3 + x^4 + x^5 + x^6).$$

We can multiply this out by choosing one term from each factor in all possible ways. If we then collect like terms, the coefficient of x^k will be the number of ways to choose one term from each factor so that the exponents of the terms add up to k . This is precisely the number of solutions to $x_1 + x_2 + x_3 + x_4 = k$, where $0 \leq x_1 \leq 2$, $0 \leq x_2 \leq 5$, $0 \leq x_3 \leq 5$, $2 \leq x_4 \leq 6$. Thus, the answer to the problem is the coefficient of x^{17} . With the help of a computer algebra system we get

$$\begin{aligned} & (1 + x + x^2)(1 + x + x^2 + x^3 + x^4 + x^5)^2(x^2 + x^3 + x^4 + x^5 + x^6) \\ &= x^{18} + 4x^{17} + 10x^{16} + 19x^{15} + 31x^{14} + 45x^{13} + 58x^{12} + 67x^{11} + 70x^{10} \\ & \quad + 67x^9 + 58x^8 + 45x^7 + 31x^6 + 19x^5 + 10x^4 + 4x^3 + x^2, \end{aligned}$$

so the answer is 4. □

EXAMPLE 3.1.4 Find the generating function for the number of solutions to $x_1 + x_2 + x_3 + x_4 = k$, where $0 \leq x_1 \leq \infty$, $0 \leq x_2 \leq 5$, $0 \leq x_3 \leq 5$, $2 \leq x_4 \leq 6$.

This is just like the previous example except that x_1 is not bounded above. The generating function is thus

$$\begin{aligned} f(x) &= (1 + x + x^2 + \cdots)(1 + x + x^2 + x^3 + x^4 + x^5)^2(x^2 + x^3 + x^4 + x^5 + x^6) \\ &= (1 - x)^{-1}(1 + x + x^2 + x^3 + x^4 + x^5)^2(x^2 + x^3 + x^4 + x^5 + x^6) \\ &= \frac{(1 + x + x^2 + x^3 + x^4 + x^5)^2(x^2 + x^3 + x^4 + x^5 + x^6)}{1 - x}. \end{aligned}$$

Note that $(1 - x)^{-1} = (1 + x + x^2 + \cdots)$ is the familiar geometric series from calculus; alternately, we could use example 3.1.2. Unlike the function in the previous example, this function has an infinite expansion:

$$\begin{aligned} f(x) &= x^2 + 4x^3 + 10x^4 + 20x^5 + 35x^6 + 55x^7 + 78x^8 \\ &\quad + 102x^9 + 125x^{10} + 145x^{11} + 160x^{12} + 170x^{13} + 176x^{14} \\ &\quad + 179x^{15} + 180x^{16} + 180x^{17} + 180x^{18} + 180x^{19} + 180x^{20} + \cdots. \end{aligned}$$

EXAMPLE 12

Use generating functions to determine the number of ways to insert tokens worth \$1, \$2, and \$5 into a vending machine to pay for an item that costs r dollars in both the cases when the order in which the tokens are inserted does not matter and when the order does matter. (For example, there are two ways to pay for an item that costs \$3 when the order in which the tokens are inserted does not matter: inserting three \$1 tokens or one \$1 token and a \$2 token. When the order matters, there are three ways: inserting three \$1 tokens, inserting a \$1 token and then a \$2 token, or inserting a \$2 token and then a \$1 token.)

Solution: Consider the case when the order in which the tokens are inserted does not matter. Here, all we care about is the number of each token used to produce a total of r dollars. Because we can use any number of \$1 tokens, any number of \$2 tokens, and any number of \$5 tokens, the answer is the coefficient of x^r in the generating function

$$(1 + x + x^2 + x^3 + \cdots)(1 + x^2 + x^4 + x^6 + \cdots)(1 + x^5 + x^{10} + x^{15} + \cdots).$$

(The first factor in this product represents the \$1 tokens used, the second the \$2 tokens used, and the third the \$5 tokens used.) For example, the number of ways to pay for an item costing \$7 using \$1, \$2, and \$5 tokens is given by the coefficient of x^7 in this expansion, which equals 6.

When the order in which the tokens are inserted matters, the number of ways to insert exactly n tokens to produce a total of r dollars is the coefficient of x^r in

$$(x + x^2 + x^5)^n,$$

because each of the r tokens may be a \$1 token, a \$2 token, or a \$5 token. Because any number of tokens may be inserted, the number of ways to produce r dollars using \$1, \$2, or \$5 tokens, when the order in which the tokens are inserted matters, is the coefficient of x^r in

$$\begin{aligned} 1 + (x + x^2 + x^5) + (x + x^2 + x^5)^2 + \cdots &= \frac{1}{1 - (x + x^2 + x^5)} \\ &= \frac{1}{1 - x - x^2 - x^5}, \end{aligned}$$

where we have added the number of ways to insert 0 tokens, 1 token, 2 tokens, 3 tokens, and so on, and where we have used the identity $1/(1 - x) = 1 + x + x^2 + \cdots$ with x replaced with $x + x^2 + x^5$. For example, the number of ways to pay for an item costing \$7 using \$1, \$2, and \$5 tokens, when the order in which the tokens are used matters, is the coefficient of x^7 in this expansion, which equals 26. [Hint: To see that this coefficient equals 26 requires the addition of the coefficients of x^7 in the expansions $(x + x^2 + x^5)^k$ for $2 \leq k \leq 7$. This can be done by hand with considerable computation, or a computer algebra system can be used.] ◀



EXAMPLE 13 Use generating functions to find the number of k -combinations of a set with n elements. Assume that the Binomial Theorem has already been established.

Solution: Each of the n elements in the set contributes the term $(1 + x)$ to the generating function $f(x) = \sum_{k=0}^n a_k x^k$. Here $f(x)$ is the generating function for $\{a_k\}$, where a_k represents the number of k -combinations of a set with n elements. Hence,

$$f(x) = (1 + x)^n.$$

But by the Binomial Theorem, we have

$$f(x) = \sum_{k=0}^n \binom{n}{k} x^k,$$

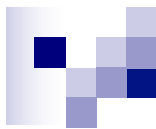
where

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Hence, $C(n, k)$, the number of k -combinations of a set with n elements, is

$$\frac{n!}{k!(n-k)!}.$$





EXAMPLE 14 Use generating functions to find the number of r -combinations from a set with n elements when repetition of elements is allowed.

Solution: Let $G(x)$ be the generating function for the sequence $\{a_r\}$, where a_r equals the number of r -combinations of a set with n elements with repetitions allowed. That is, $G(x) = \sum_{r=0}^{\infty} a_r x^r$. Because we can select any number of a particular member of the set with n elements when we form an r -combination with repetition allowed, each of the n elements contributes $(1 + x + x^2 + x^3 + \cdots)$ to a product expansion for $G(x)$. Each element contributes this factor because it may be selected zero times, one time, two times, three times, and so on, when an r -combination is formed (with a total of r elements selected). Because there are n elements in the set and each contributes this same factor to $G(x)$, we have

$$G(x) = (1 + x + x^2 + \cdots)^n.$$

As long as $|x| < 1$, we have $1 + x + x^2 + \cdots = 1/(1 - x)$, so

$$G(x) = 1/(1 - x)^n = (1 - x)^{-n}.$$

Applying the extended Binomial Theorem (Theorem 2), it follows that

$$(1 - x)^{-n} = (1 + (-x))^{-n} = \sum_{r=0}^{\infty} \binom{-n}{r} (-x)^r.$$

The number of r -combinations of a set with n elements with repetitions allowed, when r is a positive integer, is the coefficient a_r of x^r in this sum. Consequently, using Example 8 we find that a_r equals

$$\begin{aligned} \binom{-n}{r} (-1)^r &= (-1)^r C(n + r - 1, r) \cdot (-1)^r \\ &= C(n + r - 1, r). \end{aligned}$$



EXAMPLE 15

Use generating functions to find the number of ways to select r objects of n different kinds if we must select at least one object of each kind.

Solution: Because we need to select at least one object of each kind, each of the n kinds of objects contributes the factor $(x + x^2 + x^3 + \cdots)$ to the generating function $G(x)$ for the sequence $\{a_r\}$, where a_r is the number of ways to select r objects of n different kinds if we need at least one object of each kind. Hence,

$$G(x) = (x + x^2 + x^3 + \cdots)^n = x^n(1 + x + x^2 + \cdots)^n = x^n/(1 - x)^n.$$

Using the extended binomial theorem and Example 8, we have

$$\begin{aligned} G(x) &= x^n/(1 - x)^n \\ &= x^n \cdot (1 - x)^{-n} \\ &= x^n \sum_{r=0}^{\infty} \binom{-n}{r} (-x)^r \\ &= x^n \sum_{r=0}^{\infty} (-1)^r C(n + r - 1, r) (-1)^r x^r \\ &= \sum_{r=0}^{\infty} C(n + r - 1, r) x^{n+r} \\ &= \sum_{t=n}^{\infty} C(t - 1, t - n) x^t \\ &= \sum_{r=n}^{\infty} C(r - 1, r - n) x^r. \end{aligned}$$

$t = n + r, \therefore r = t - n$

$t = r, \therefore \begin{cases} t - 1 = r - 1 \\ t - n = r - n \end{cases}$

We have shifted the summation in the next-to-last equality by setting $t = n + r$ so that $t = n$ when $r = 0$ and $n + r - 1 = t - 1$, and then we replaced t by r as the index of summation in the last equality to return to our original notation. Hence, there are $C(r - 1, r - n)$ ways to select r objects of n different kinds if we must select at least one object of each kind. ◀



Exponential Generating Functions

$$e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n \text{ is the generating function for the sequence } \frac{1}{0!}, \frac{1}{1!}, \frac{1}{2!}, \frac{1}{3!}, \dots$$

$$\Downarrow$$
$$e^x = \sum_{n=0}^{\infty} 1 \cdot \frac{x^n}{n!}$$

considering the $n!$ to be part of the expression $x^n/n!$, we might think of this same function as generating the sequence $1, 1, 1, \dots$, interpreting 1 as the coefficient of $x^n/n!$. This is not a very interesting sequence, of course, but this idea can often prove fruitful. If

$$f(x) = \sum_{n=0}^{\infty} a_n \frac{x^n}{n!},$$

we say that $f(x)$ is the **exponential generating function** for $a_0, a_1, a_2, \dots$



Exponential Generating Functions

- Find the number of permutations of a multiset can be very difficult.
- We can make it easier by introducing exponential generating function.

Let $h_0, h_1, h_2, \dots$ be a sequence.

The exponential generating function for the sequence is

$$h(x) = \frac{h_0}{0!}x^0 + \frac{h_1}{1!}x^1 + \frac{h_2}{2!}x^2 + \frac{h_3}{3!}x^3 + \dots$$

Note that if $h_i = 1$ for all i , this becomes

$$\frac{1}{0!}x^0 + \frac{1}{1!}x^1 + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \dots = e^x$$



Exponential Generating Functions

- Find the exponential generating function for the sequence $h_k = k!$.

Since $h_k = k!$, the generating function is

$$h(x) = \frac{0!}{0!}x^0 + \frac{1!}{1!}x^1 + \frac{2!}{2!}x^2 + \frac{3!}{3!}x^3 + \dots$$

$$= 1 + x + x^2 + x^3 + \dots$$

$$= \frac{1}{1-x}$$



Exponential Generating Functions

- Find the exponential generating function for h_i , where h_i is the number i -permutations of the multiset $\{4 \times a\} = \{a, a, a, a\}$.

There is only one each of the 0-, 1-, 2-, 3-, 4-permutations

0-permutations:

1-permutations: a

2-permutations: aa

3-permutations: aaa

4-permutations: $aaaa$

$\Rightarrow$ So the h_i sequences is
1, 1, 1, 1, 1, 0, 0, 0, 0, ...

giving us

$$g(x) = \frac{1}{0!}x^0 + \frac{1!}{1!}x^1 + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \frac{1}{4!}x^4 + \frac{0}{5!}x^5 + \frac{0}{6!}x^6 + \dots$$

Exponential Generating Functions

- Find the exponential generating function for h_i , where h_i is the number i -permutations of the multiset $\{\infty \times a\} = \{a, a, \dots\}$ and a appears an even number of times.

Since the only element we can choose is a , and a must appear an even number of times, then

$$h_1 = h_3 = h_5 = \dots = 0$$

and. $h_0 = h_2 = h_4 = \dots = 1$

So the exponential generating function is

$$g(x) = \frac{1}{0!}x^0 + \frac{1}{2!}x^2 + \frac{1}{4!}x^4 + \dots$$

$$\frac{e^x + e^{-x}}{2} = \frac{1}{0!}x^0 + \frac{1}{2!}x^2 + \frac{1}{4!}x^4 + \dots$$

$$e^x = \frac{1}{0!}x^0 + \frac{1}{1!}x^1 + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \frac{1}{4!}x^4 + \dots$$

$$e^{-x} = \frac{1}{0!}x^0 - \frac{1}{1!}x^1 + \frac{1}{2!}x^2 - \frac{1}{3!}x^3 + \frac{1}{4!}x^4 + \dots$$

$$e^x + e^{-x} = 2 \cdot \frac{1}{0!}x^0 + 2 \cdot \frac{1}{2!}x^2 + 2 \cdot \frac{1}{4!}x^4 + \dots$$

$$\frac{e^x + e^{-x}}{2} = \frac{1}{0!}x^0 + \frac{1}{2!}x^2 + \frac{1}{4!}x^4 + \dots$$

Exponential Generating Functions

- Find the number 3-permutations of the multiset $\{2 \times a, 3 \times b\} = \{a, a, b, b, b\}$.

e.g. There should be at least 3 kinds of multiset

$$\begin{array}{l}
 \{b, b, b\}, \text{ which number } \frac{3!}{0! 3!} \\
 \{a, b, b\}, \text{ which number } \frac{3!}{1! 2!} \\
 \{a, a, b\}, \text{ which number } \frac{3!}{2! 1!}
 \end{array} \Rightarrow \therefore \frac{3!}{0! 3!} + \frac{3!}{1! 2!} + \frac{3!}{2! 1!}$$

So the exponential generating function $h_1 = \{2 \times a\}$

$$h_1(x) = \frac{1}{0!}x^0 + \frac{1!}{1!}x^1 + \frac{1}{2!}x^2$$

So the exponential generating function $h_2 = \{3 \times b\}$

$$h_2(x) = \frac{1}{0!}x^0 + \frac{1!}{1!}x^1 + \frac{1}{2!}x^2 + \frac{1}{3!}x^3$$



Exponential Generating Functions

- Find the number 3-permutations of the multiset $\{2 \times a, 3 \times b\} = \{a, a, b, b, b\}$.

So the exponential generating function $h_1 = \{2 \times a\}$

$$h_1(x) = \frac{1}{0!}x^0 + \frac{1!}{1!}x^1 + \frac{1}{2!}x^2$$

So the exponential generating function $h_2 = \{3 \times b\}$

$$h_2(x) = \frac{1}{0!}x^0 + \frac{1!}{1!}x^1 + \frac{1}{2!}x^2 + \frac{1}{3!}x^3$$

Expanding

$$h_1(x) h_2(x) = \dots + x^3 \cdot \left(\frac{1}{0!} \cdot \frac{1}{3!} + \frac{1}{1!} \cdot \frac{1}{2!} + \frac{1}{2!} \cdot \frac{1}{1!} \right) +$$

But
$$\frac{1}{0!} \cdot \frac{1}{3!} + \frac{1}{1!} \cdot \frac{1}{2!} + \frac{1}{2!} \cdot \frac{1}{1!} = \frac{1}{3!} \cdot \left(\frac{3!}{0! 3!} + \frac{3!}{1! 2!} + \frac{3!}{2! 1!} \right)$$

Theorem

Let $h_i(x)$ be the exponential generating function for the multiset $\{n_i \cdot i\}$. Then

$$g(x) = h_1(x)h_2(x) \cdots h_k(x)$$

will be the exponential generating function for

$$\{n_1 \cdot 1, n_2 \cdot 2, \dots, n_k \cdot k\}$$

A Proof

Let

$$g(x) = \frac{g_0}{0!} + \frac{g_1}{1!}x + \frac{g_2}{2!}x^2 + \dots$$

The x^n coefficient of $g(x)$ will be

$$\sum \frac{1}{p_1!p_2! \cdots p_k!}$$

over all $p_1, p_2, \dots, p_k$ where

$$n = p_1 + p_2 + \dots + p_k$$

and a n -permutation with p_1 1s, p_2 2s, $\dots$, p_k k s is possible.

Consequently

$$g_n = \sum \frac{n!}{p_1!p_2! \cdots p_k!}$$

Consequently $g(x) = h_1(x)h_2(x) \cdots h_k(x)$ is an exponential generating function for the number of permutations of the multiset.

Meanwhile, the number of ways of forming a n -permutation from the multiset will be the sum of all possible

$$\frac{n!}{p_1!p_2! \cdots p_k!}$$

where

$$n = p_1 + p_2 + \dots + p_k$$

and a permutation with p_1 1s, p_2 2s, $\dots$, p_k k s is possible.

But this is g_n .



Partitions of Integers

DEFINITION 3.3.1 A **partition** of a positive integer n is a multiset of positive integers that sum to n . We denote the number of partitions of n by p_n . $\square$

Typically a partition is written as a sum, not explicitly as a multiset. Using the usual convention that an empty sum is 0, we say that $p_0 = 1$.

EXAMPLE 3.3.2 The partitions of 5 are

5
4 + 1
3 + 2
3 + 1 + 1
2 + 2 + 1
2 + 1 + 1 + 1
1 + 1 + 1 + 1 + 1.

Thus $p_5 = 7$. $\square$



Partitions of Integers

There is no simple formula for p_n , but it is not hard to find a generating function for them. As with some previous examples, we seek a product of factors so that when the factors are multiplied out, the coefficient of x^n is p_n . We would like each x^n term to represent a single partition, before like terms are collected. A partition is uniquely described by the number of 1s, number of 2s, and so on, that is, by the repetition numbers of the multiset. We devote one factor to each integer:

$$(1 + x + x^2 + x^3 + \cdots)(1 + x^2 + x^4 + x^6 + \cdots) \cdots (1 + x^k + x^{2k} + x^{3k} + \cdots) \cdots = \prod_{k=1}^{\infty} \sum_{i=0}^{\infty} x^{ik}.$$

When this product is expanded, we pick one term from each factor in all possible ways, with the further condition that we only pick a finite number of “non-1” terms. For example, if we pick x^3 from the first factor, x^3 from the third factor, x^{15} from the fifth factor, and 1s from all other factors, we get x^{21} . In the context of the product, this represents $3 \cdot 1 + 1 \cdot 3 + 3 \cdot 5$, corresponding to the partition $1 + 1 + 1 + 3 + 5 + 5 + 5$, that is, three 1s, one 3, and three 5s. Each factor is a geometric series; the k th factor is

$$1 + x^k + (x^k)^2 + (x^k)^3 + \cdots = \frac{1}{1 - x^k},$$

so the generating function can be written

$$\prod_{k=1}^{\infty} \frac{1}{1 - x^k}.$$



Partitions of Integers

EXAMPLE 3.3.3 Find p_8 .

We expand

$$\begin{aligned} & (1 + x + x^2 + x^3 + x^4 + x^5 + x^6 + x^7 + x^8)(1 + x^2 + x^4 + x^6 + x^8)(1 + x^3 + x^6) \\ & (1 + x^4 + x^8)(1 + x^5)(1 + x^6)(1 + x^7)(1 + x^8) \\ & = 1 + x + 2x^2 + 3x^3 + 5x^4 + 7x^5 + 11x^6 + 15x^7 + 22x^8 + \cdots + x^{56}, \end{aligned}$$

so $p_8 = 22$. Note that all of the coefficients prior to this are also correct, but the following coefficients are not necessarily the corresponding partition numbers. $\square$

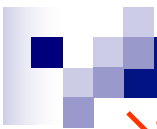
EXAMPLE 3.3.4 For $n = 6$, the partitions into distinct parts are

$$6, 5 + 1, 4 + 2, 3 + 2 + 1,$$

so $p_d(6) = 4$, and the partitions into odd parts are

$$5 + 1, 3 + 3, 3 + 1 + 1 + 1, 1 + 1 + 1 + 1 + 1 + 1,$$

so $p_o(6) = 4$. $\square$



✕Using Generating Functions to solve Recurrence Relations.

Example 18.

Solving the recurrence relation $a_k = 3a_{k-1}$ for $k=1,2,3,\dots$ and initial condition $a_0 = 2$.

Sol :

另法：(by 8.2公式)

$$r - 3 = 0 \Rightarrow r = 3 \Rightarrow a_n = \alpha \cdot 3^n$$

$$\because a_0 = 2 = \alpha$$

$$\therefore a_n = 2 \cdot 3^n$$

Let $G(x) = a_0 + a_1x + a_2x^2 + \dots = \sum_{k=0}^{\infty} a_k x^k$ be the generating function for $\{a_k\}$.

First note that $a_k = 3a_{k-1}$

$$\Rightarrow \sum_{k=1}^{\infty} a_k x^k = 3 \sum_{k=1}^{\infty} a_{k-1} x^k = 3x \sum_{k=1}^{\infty} a_{k-1} x^{k-1} = 3x \sum_{k=0}^{\infty} a_k x^k$$

從 $k=1$ 開始，以避免 a_{k-1} 變成 a_{-1}

$$\Rightarrow G(x) - \underline{a_0} = 3x \cdot G(x)$$

$$\because a_0 = 2 \Rightarrow G(x) - 3x \cdot G(x) = G(x)(1-3x) = 2$$

$$\therefore G(x) = \frac{2}{1-3x} = 2 \cdot \sum_{k=0}^{\infty} (3x)^k = \sum_{k=0}^{\infty} 2 \cdot 3^k \cdot x^k$$

$$\therefore a_k = 2 \cdot 3^k$$

Example 19

Solving $a_k = 8a_{k-1} + 10^{k-1}$ for $k = 1, 2, 3, \dots$ and initial condition $a_1 = 9$.

Sol : Let $a_0 = 1$ (為計算方便而假設).

Let $G(x) = a_0 + a_1x + a_2x^2 + \dots = \sum_{k=0}^{\infty} a_k x^k$
be the generating function for $\{a_k\}$.

$$\begin{aligned} G(x) - 1 &= \sum_{k=1}^{\infty} a_k x^k = \sum_{k=1}^{\infty} (8a_{k-1} + 10^{k-1}) x^k \\ &= 8 \sum_{k=1}^{\infty} a_{k-1} x^k + \sum_{k=1}^{\infty} 10^{k-1} x^k = 8x \sum_{k=0}^{\infty} a_k x^k + x \sum_{k=0}^{\infty} 10^k x^k \\ &= 8xG(x) + \frac{x}{1-10x} \end{aligned}$$


$$(1-8x)G(x) = 1 + \frac{x}{1-10x} = \frac{1-9x}{1-10x}$$

$$G(x) = \frac{1-9x}{(1-10x)(1-8x)} = \frac{1}{2} \left(\frac{1}{1-10x} + \frac{1}{1-8x} \right)$$

$$= \frac{1}{2} \left(\sum_{k=0}^{\infty} 10^k x^k + \sum_{k=0}^{\infty} 8^k x^k \right) = \sum_{k=0}^{\infty} \frac{1}{2} (10^k + 8^k) x^k$$

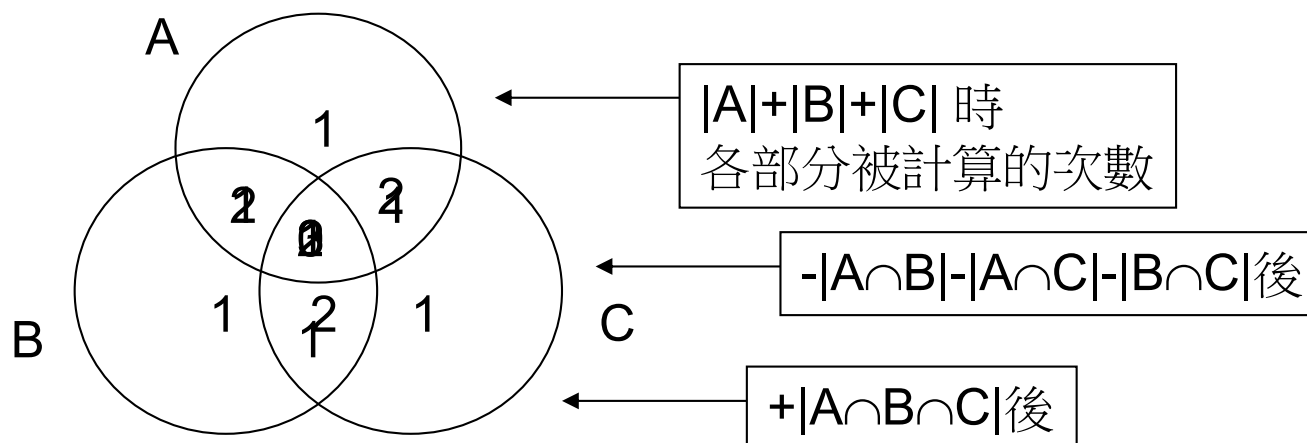
$$\therefore a_k = (10^k + 8^k)/2$$

8.5 Inclusion-Exclusion 排容原理

A, B, C, D : sets

(1). $|A \cup B| = |A| + |B| - |A \cap B|$

(2). $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$



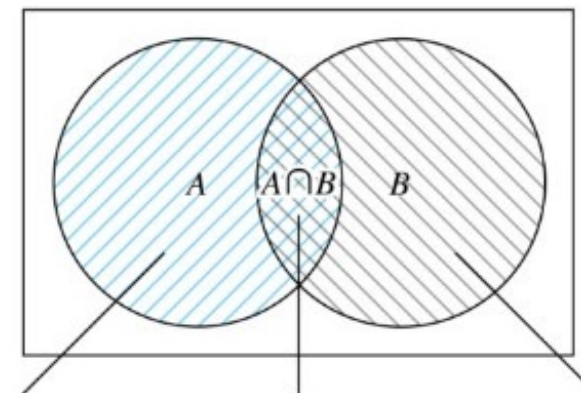
(3). $|A \cup B \cup C \cup D| = ?$

Two Finite Sets

Example: In a discrete mathematics class every student is a major in computer science or mathematics or both. The number of students having computer science as a major (possibly along with mathematics) is 25; the number of students having mathematics as a major (possibly along with computer science) is 13; and the number of students majoring in both computer science and mathematics is 8. How many students are in the class?

Solution: $|A \cup B| = |A| + |B| - |A \cap B|$
 $= 25 + 13 - 8 = 30$

$$|A \cup B| = |A| + |B| - |A \cap B| = 25 + 13 - 8 = 30$$



$$|A|=25$$

$$|A \cap B|=8$$

$$|B|=13$$

[Jump to long description](#)

FIG

$$|A \cup B| = |A| + |B| - |A \cap B| = 142 + 90 - 12 = 220$$

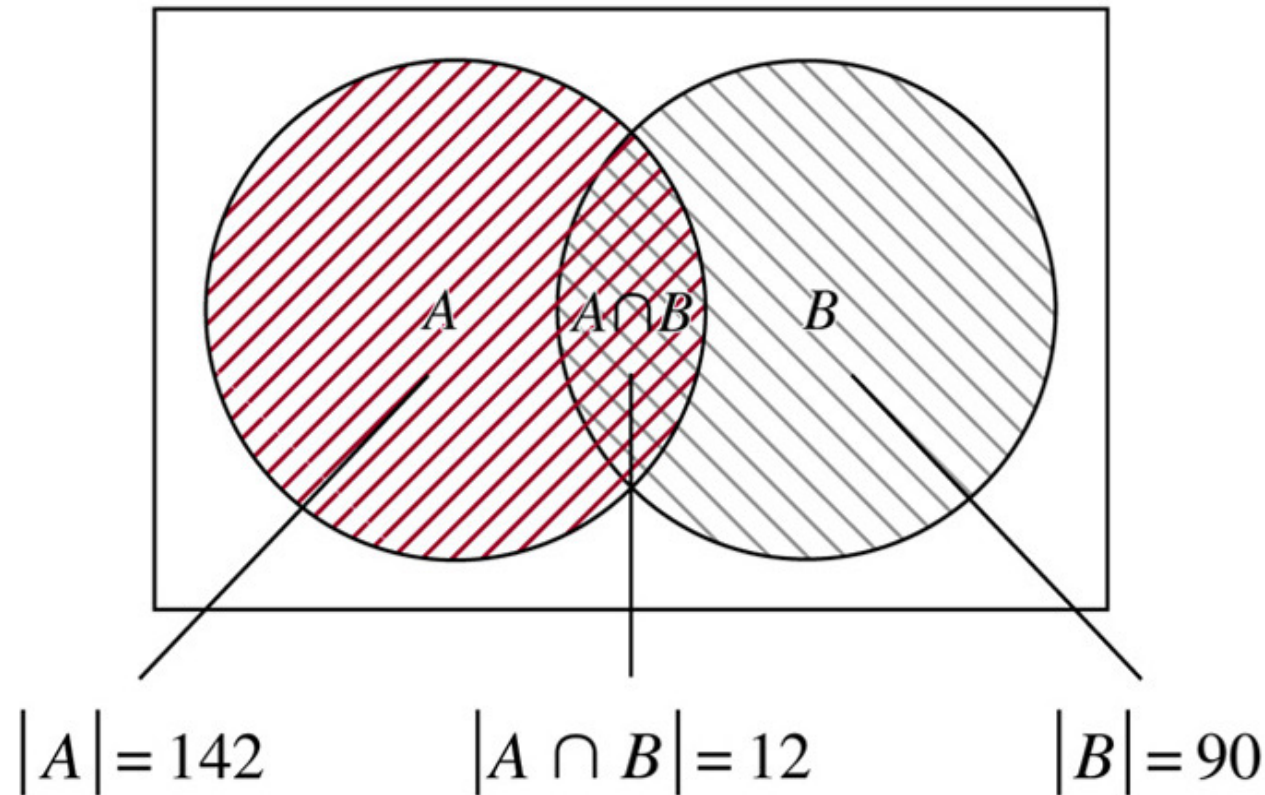


FIGURE 2 The Set of Positive Integers Not Exceeding 1000 Divisible by Either 7 or 11.



EXAMPLE 3

Suppose that there are 1807 freshmen at your school. Of these, 453 are taking a course in computer science, 567 are taking a course in mathematics, and 299 are taking courses in both computer science and mathematics. How many are not taking a course either in computer science or in mathematics?

Solution: To find the number of freshmen who are not taking a course in either mathematics or computer science, subtract the number that are taking a course in either of these subjects from the total number of freshmen. Let A be the set of all freshmen taking a course in computer science, and let B be the set of all freshmen taking a course in mathematics. It follows that $|A| = 453$, $|B| = 567$, and $|A \cap B| = 299$. The number of freshmen taking a course in either computer science or mathematics is

$$|A \cup B| = |A| + |B| - |A \cap B| = 453 + 567 - 299 = 721.$$

Consequently, there are $1807 - 721 = 1086$ freshmen who are not taking a course in computer science or mathematics. 

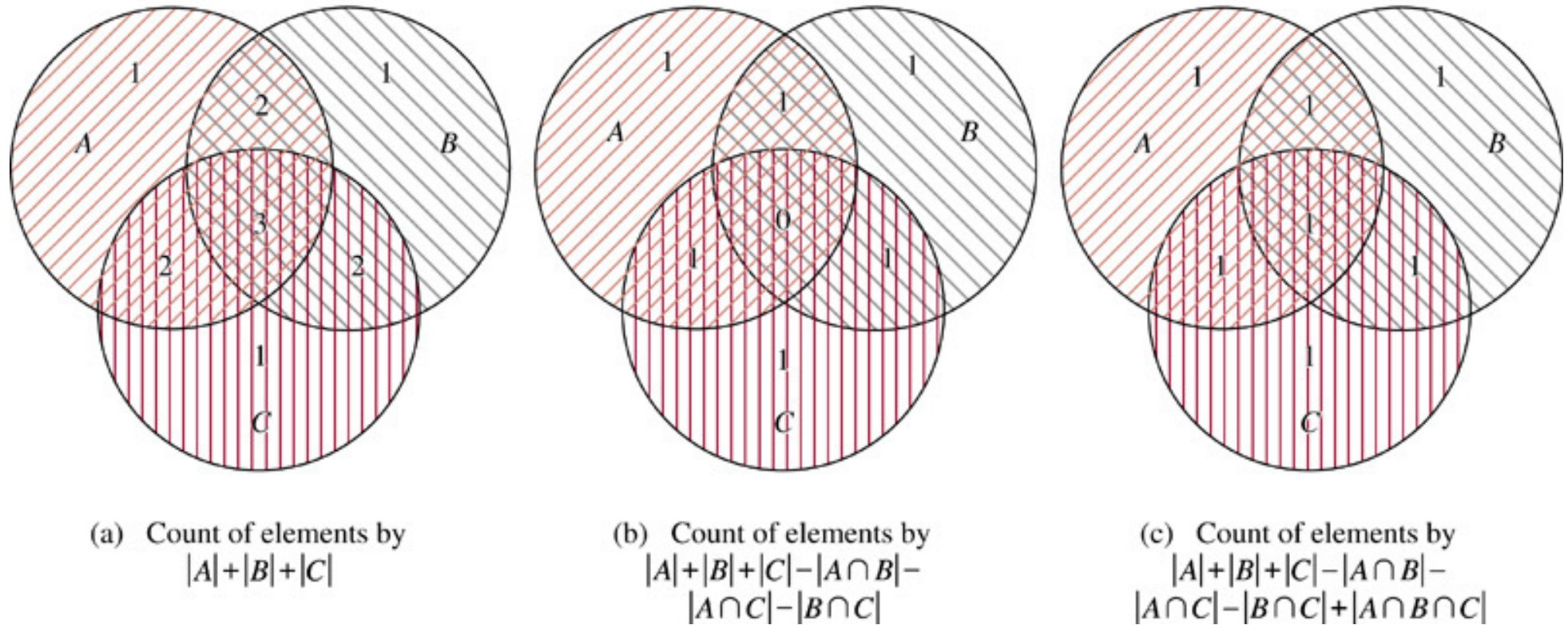


FIGURE 3 Finding a Formula for the Number of Elements in the Union of Three Sets.



Three Finite Sets

Example: A total of 1232 students have taken a course in Spanish, 879 have taken a course in French, and 114 have taken a course in Russian. Further, 103 have taken courses in both Spanish and French, 23 have taken courses in both Spanish and Russian, and 14 have taken courses in both French and Russian. If 2092 students have taken a course in at least one of Spanish French and Russian, how many students have taken a course in all 3 languages.

Solution: Let S be the set of students who have taken a course in Spanish, F the set of students who have taken a course in French, and R the set of students who have taken a course in Russian. Then, we have $|S| = 1232$, $|F| = 879$, $|R| = 114$, $|S \cap F| = 103$, $|S \cap R| = 23$, $|F \cap R| = 14$, and $|S \cup F \cup R| = 2092$.

Using the equation

$$|S \cup F \cup R| = |S| + |F| + |R| - |S \cap F| - |S \cap R| - |F \cap R| + |S \cap F \cap R|,$$

we obtain $2092 = 1232 + 879 + 114 - 103 - 23 - 14 + |S \cap F \cap R|$.

Solving for $|S \cap F \cap R|$ yields 7.

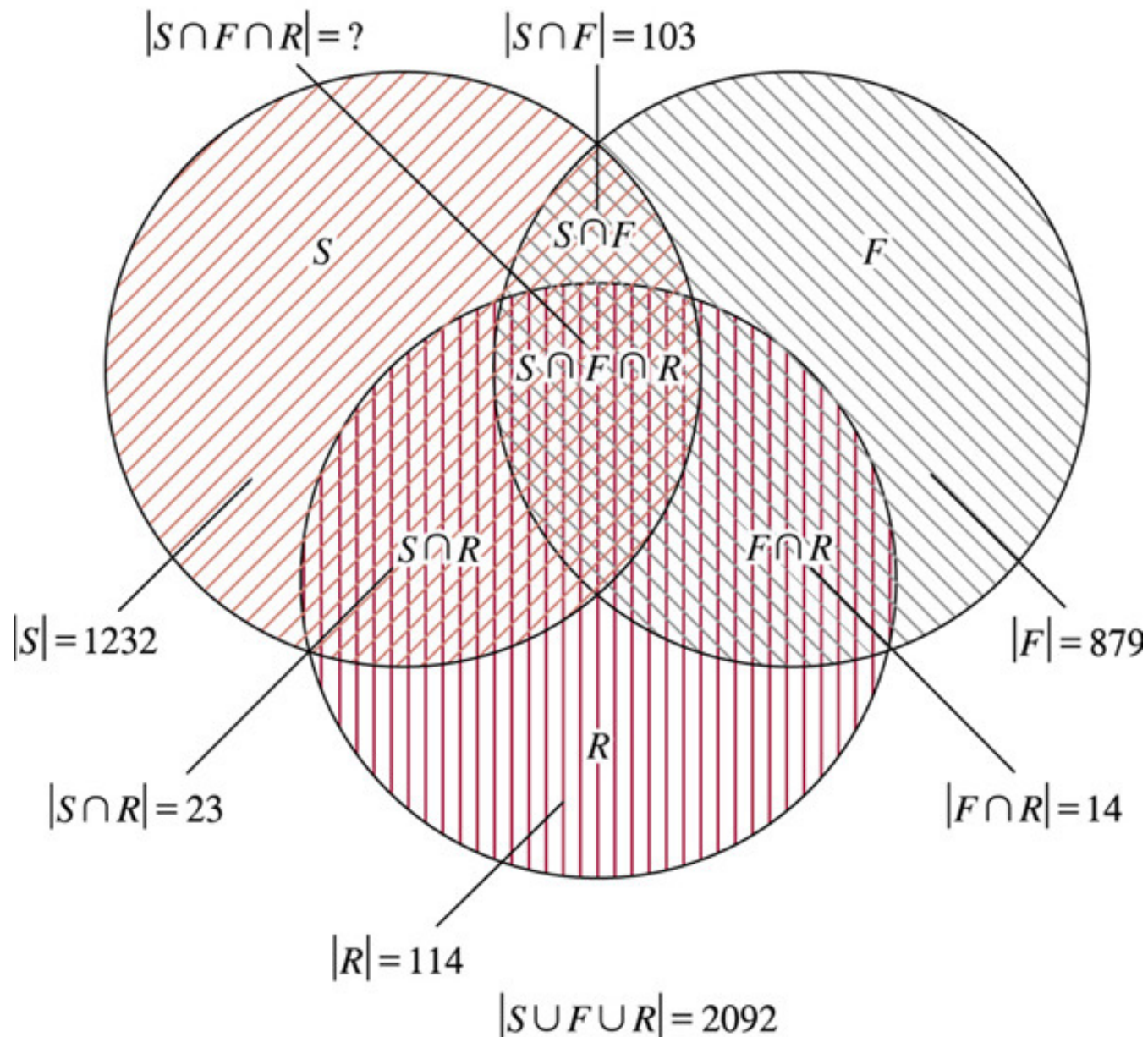


FIGURE 4

The Set of Students Who Have Taken Courses in Spanish, French, and Russian.



Theorem 1.

$A_1, A_2, \dots, A_n$: sets

$$\begin{aligned} |A_1 \cup A_2 \cup \dots \cup A_n| &= \sum_{i=1}^n |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| \\ &+ \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| - \dots + (-1)^{n+1} \cdot |A_1 \cap A_2 \cap \dots \cap A_n| \end{aligned}$$



The Principle of Inclusion-Exclusion

Proof: An element in the union is counted exactly once in the right-hand side of the equation.

Consider an element a that is a member of r of the sets $A_1, \dots, A_n$ where $1 \leq r \leq n$.

- It is counted $C(r, 1)$ times by $\sum |A_i|$
- It is counted $C(r, 2)$ times by $\sum |A_i \cap A_j|$
- In general, it is counted $C(r, m)$ times by the summation of m of the sets A_i .



The Principle of Inclusion-Exclusion

Thus the element is counted exactly

$$C(r,1) - C(r,2) + C(r,3) - \dots + (-1)^{r+1} C(r,r) = 0.$$

times by the right hand side of the equation.

By Corollary 2 of Section 6.4, we have

$$C(r,0) - C(r,1) + C(r,2) - \dots + (-1)^r C(r,r) = 0.$$

Hence,

$$1 = C(r,0) = C(r,1) - C(r,2) + \dots + (-1)^{r+1} C(r,r).$$

8.6 Applications of Inclusion and Exclusion

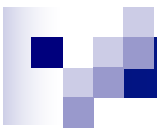
Example 1. How many solutions does $x_1 + x_2 + x_3 = 11$ have, where x_1, x_2, x_3 are nonnegative integers with $x_1 \leq 3$, $x_2 \leq 4$, and $x_3 \leq 6$?

Sol :

Let a solution have property P_1 if $x_1 \geq 4$, property P_2 if $x_2 \geq 5$, and property P_3 if $x_3 \geq 7$.

$$N(P_1'P_2'P_3') = N - N(P_1) - N(P_2) - N(P_3) + N(P_1P_2) + N(P_2P_3) + N(P_1P_3) - N(P_1P_2P_3)$$

$$= \binom{11+2}{11} - \binom{7+2}{7} - \binom{6+2}{6} - \binom{4+2}{4} + \binom{2+2}{2} + \binom{0+2}{0} = 6$$



From the inclusion–exclusion principle, we see that

$$\begin{aligned} N(P'_1 P'_2 \dots P'_n) = N &- \sum_{1 \leq i \leq n} N(P_i) + \sum_{1 \leq i < j \leq n} N(P_i P_j) \\ &- \sum_{1 \leq i < j < k \leq n} N(P_i P_j P_k) + \dots + (-1)^n N(P_1 P_2 \dots P_n). \end{aligned}$$

EXAMPLE 1 How many solutions does

$$x_1 + x_2 + x_3 = 11$$

have, where x_1 , x_2 , and x_3 are nonnegative integers with $x_1 \leq 3$, $x_2 \leq 4$, and $x_3 \leq 6$?

Solution: To apply the principle of inclusion–exclusion, let a solution have property P_1 if $x_1 > 3$, property P_2 if $x_2 > 4$, and property P_3 if $x_3 > 6$. The number of solutions satisfying the inequalities $x_1 \leq 3$, $x_2 \leq 4$, and $x_3 \leq 6$ is

$$\begin{aligned} N(P'_1 P'_2 P'_3) &= N - N(P_1) - N(P_2) - N(P_3) + N(P_1 P_2) \\ &\quad + N(P_1 P_3) + N(P_2 P_3) - N(P_1 P_2 P_3). \end{aligned}$$

Using the same techniques as in Example 5 of Section 5.5, it follows that

- $N = \text{total number of solutions} = C(3 + 11 - 1, 11) = 78$,
- $N(P_1) = (\text{number of solutions with } x_1 \geq 4) = C(3 + 7 - 1, 7) = C(9, 7) = 36$,
- $N(P_2) = (\text{number of solutions with } x_2 \geq 5) = C(3 + 6 - 1, 6) = C(8, 6) = 28$,
- $N(P_3) = (\text{number of solutions with } x_3 \geq 7) = C(3 + 4 - 1, 4) = C(6, 4) = 15$,
- $N(P_1 P_2) = (\text{number of solutions with } x_1 \geq 4 \text{ and } x_2 \geq 5) = C(3 + 2 - 1, 2) = C(4, 2) = 6$,
- $N(P_1 P_3) = (\text{number of solutions with } x_1 \geq 4 \text{ and } x_3 \geq 7) = C(3 + 0 - 1, 0) = 1$,
- $N(P_2 P_3) = (\text{number of solutions with } x_2 \geq 5 \text{ and } x_3 \geq 7) = 0$,
- $N(P_1 P_2 P_3) = (\text{number of solutions with } x_1 \geq 4, x_2 \geq 5, \text{ and } x_3 \geq 7) = 0$.

Inserting these quantities into the formula for $N(P'_1 P'_2 P'_3)$ shows that the number of solutions with $x_1 \leq 3$, $x_2 \leq 4$, and $x_3 \leq 6$ equals

$$N(P'_1 P'_2 P'_3) = 78 - 36 - 28 - 15 + 6 + 1 + 0 - 0 = 6.$$



TABLE 1 The Sieve of Eratosthenes.

Integers divisible by 2 other than 2 receive an underline.

1	2	3	<u>4</u>	5	<u>6</u>	7	<u>8</u>	9	<u>10</u>
11	<u>12</u>	13	<u>14</u>	15	<u>16</u>	17	<u>18</u>	19	<u>20</u>
21	<u>22</u>	23	<u>24</u>	25	<u>26</u>	27	<u>28</u>	29	<u>30</u>
31	<u>32</u>	33	<u>34</u>	35	<u>36</u>	37	<u>38</u>	39	<u>40</u>
41	<u>42</u>	43	<u>44</u>	45	<u>46</u>	47	<u>48</u>	49	<u>50</u>
51	<u>52</u>	53	<u>54</u>	55	<u>56</u>	57	<u>58</u>	59	<u>60</u>
61	<u>62</u>	63	<u>64</u>	65	<u>66</u>	67	<u>68</u>	69	<u>70</u>
71	<u>72</u>	73	<u>74</u>	75	<u>76</u>	77	<u>78</u>	79	<u>80</u>
81	<u>82</u>	83	<u>84</u>	85	<u>86</u>	87	<u>88</u>	89	<u>90</u>
91	<u>92</u>	93	<u>94</u>	95	<u>96</u>	97	<u>98</u>	99	<u>100</u>

Integers divisible by 3 other than 3 receive an underline.

1	2	3	<u>4</u>	5	<u>6</u>	7	8	<u>9</u>	<u>10</u>
11	<u>12</u>	13	<u>14</u>	<u>15</u>	<u>16</u>	17	<u>18</u>	19	<u>20</u>
<u>21</u>	<u>22</u>	23	<u>24</u>	25	<u>26</u>	<u>27</u>	<u>28</u>	29	<u>30</u>
31	<u>32</u>	<u>33</u>	<u>34</u>	35	<u>36</u>	37	<u>38</u>	<u>39</u>	<u>40</u>
41	<u>42</u>	43	<u>44</u>	<u>45</u>	<u>46</u>	47	<u>48</u>	49	<u>50</u>
<u>51</u>	<u>52</u>	53	<u>54</u>	55	<u>56</u>	<u>57</u>	<u>58</u>	59	<u>60</u>
61	<u>62</u>	<u>63</u>	<u>64</u>	65	<u>66</u>	67	<u>68</u>	<u>69</u>	<u>70</u>
71	<u>72</u>	73	<u>74</u>	<u>75</u>	<u>76</u>	77	<u>78</u>	79	<u>80</u>
<u>81</u>	<u>82</u>	83	<u>84</u>	85	<u>86</u>	<u>87</u>	<u>88</u>	89	<u>90</u>
91	<u>92</u>	<u>93</u>	<u>94</u>	95	<u>96</u>	97	<u>98</u>	<u>99</u>	<u>100</u>

Integers divisible by 5 other than 5 receive an underline.

1	2	3	<u>4</u>	5	<u>6</u>	7	<u>8</u>	<u>9</u>	<u>10</u>
11	<u>12</u>	13	<u>14</u>	<u>15</u>	<u>16</u>	17	<u>18</u>	19	<u>20</u>
<u>21</u>	<u>22</u>	23	<u>24</u>	<u>25</u>	<u>26</u>	27	<u>28</u>	29	<u>30</u>
31	<u>32</u>	<u>33</u>	<u>34</u>	<u>35</u>	<u>36</u>	37	<u>38</u>	<u>39</u>	<u>40</u>
41	<u>42</u>	43	<u>44</u>	<u>45</u>	<u>46</u>	47	<u>48</u>	49	<u>50</u>
<u>51</u>	<u>52</u>	53	<u>54</u>	<u>55</u>	<u>56</u>	<u>57</u>	<u>58</u>	59	<u>60</u>
61	<u>62</u>	<u>63</u>	<u>64</u>	<u>65</u>	<u>66</u>	67	<u>68</u>	<u>69</u>	<u>70</u>
71	<u>72</u>	73	<u>74</u>	<u>75</u>	<u>76</u>	77	<u>78</u>	79	<u>80</u>
<u>81</u>	<u>82</u>	83	<u>84</u>	<u>85</u>	<u>86</u>	<u>87</u>	<u>88</u>	89	<u>90</u>
91	<u>92</u>	<u>93</u>	<u>94</u>	<u>95</u>	<u>96</u>	97	<u>98</u>	<u>99</u>	<u>100</u>

Integers divisible by 7 other than 7 receive an underline; integers in color are prime.

1	2	3	4	5	<u>6</u>	7	8	<u>9</u>	<u>10</u>
11	<u>12</u>	13	<u>14</u>	<u>15</u>	<u>16</u>	17	<u>18</u>	19	<u>20</u>
<u>21</u>	<u>22</u>	23	<u>24</u>	25	26	27	<u>28</u>	29	<u>30</u>
31	<u>32</u>	<u>33</u>	<u>34</u>	<u>35</u>	<u>36</u>	37	<u>38</u>	<u>39</u>	<u>40</u>
41	<u>42</u>	43	<u>44</u>	<u>45</u>	<u>46</u>	47	<u>48</u>	<u>49</u>	<u>50</u>
51	52	53	<u>54</u>	55	<u>56</u>	57	<u>58</u>	59	<u>60</u>
61	<u>62</u>	<u>63</u>	<u>64</u>	65	<u>66</u>	67	<u>68</u>	<u>69</u>	<u>70</u>
71	<u>72</u>	73	<u>74</u>	<u>75</u>	<u>76</u>	77	<u>78</u>	79	<u>80</u>
81	<u>82</u>	83	<u>84</u>	85	<u>86</u>	<u>87</u>	<u>88</u>	89	<u>90</u>
91	<u>92</u>	<u>93</u>	<u>94</u>	95	<u>96</u>	97	<u>98</u>	<u>99</u>	<u>100</u>

$$4 + N(P'_1 P'_2 P'_3 P'_4).$$

Because there are 99 positive integers greater than 1 and not exceeding 100, the principle of inclusion–exclusion shows that

$$\begin{aligned} N(P'_1 P'_2 P'_3 P'_4) &= 99 - N(P_1) - N(P_2) - N(P_3) - N(P_4) \\ &\quad + N(P_1 P_2) + N(P_1 P_3) + N(P_1 P_4) + N(P_2 P_3) + N(P_2 P_4) + N(P_3 P_4) \\ &\quad - N(P_1 P_2 P_3) - N(P_1 P_2 P_4) - N(P_1 P_3 P_4) - N(P_2 P_3 P_4) \\ &\quad + N(P_1 P_2 P_3 P_4). \end{aligned}$$

The number of integers not exceeding 100 (and greater than 1) that are divisible by all the primes in a subset of $\{2, 3, 5, 7\}$ is $\lfloor 100/N \rfloor$, where N is the product of the primes in this subset. (This follows because any two of these primes have no common factor.) Consequently,

$$\begin{aligned} N(P'_1 P'_2 P'_3 P'_4) &= 99 - \left\lfloor \frac{100}{2} \right\rfloor - \left\lfloor \frac{100}{3} \right\rfloor - \left\lfloor \frac{100}{5} \right\rfloor - \left\lfloor \frac{100}{7} \right\rfloor \\ &\quad + \left\lfloor \frac{100}{2 \cdot 3} \right\rfloor + \left\lfloor \frac{100}{2 \cdot 5} \right\rfloor + \left\lfloor \frac{100}{2 \cdot 7} \right\rfloor + \left\lfloor \frac{100}{3 \cdot 5} \right\rfloor + \left\lfloor \frac{100}{3 \cdot 7} \right\rfloor + \left\lfloor \frac{100}{5 \cdot 7} \right\rfloor \\ &\quad - \left\lfloor \frac{100}{2 \cdot 3 \cdot 5} \right\rfloor - \left\lfloor \frac{100}{2 \cdot 3 \cdot 7} \right\rfloor - \left\lfloor \frac{100}{2 \cdot 5 \cdot 7} \right\rfloor - \left\lfloor \frac{100}{3 \cdot 5 \cdot 7} \right\rfloor + \left\lfloor \frac{100}{2 \cdot 3 \cdot 5 \cdot 7} \right\rfloor \\ &= 99 - 50 - 33 - 20 - 14 + 16 + 10 + 7 + 6 + 4 + 2 - 3 - 2 - 1 - 0 + 0 \\ &= 21. \end{aligned}$$

Hence, there are $4 + 21 = 25$ primes not exceeding 100.

The **Sieve of Eratosthenes** is used to find all primes not exceeding a specified positive integer. For instance, the following procedure is used to find the primes not exceeding 100. First the integers that are divisible by 2, other than 2, are deleted. Because 3 is the first integer greater

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- **Example .** How many permutations of the letters

M, A, T, H, I, S, F, U, N

are there such that none of the words MATH, IS, and FUN occur as consecutive letters? (Thus, for instance, the permutation MATHISFUN is not allowed, nor are the permutations INUMATHSF and ISMATHFUN.)

Sol :

$$\begin{aligned} |\overline{A_1} \cap \overline{A_2} \cap \overline{A_3}| &= |S| - (|A_1| + |A_2| + |A_3|) + \\ &\quad (|A_1 \cap A_2| + |A_1 \cap A_3| + |A_2 \cap A_3|) \\ &\quad - |A_1 \cap A_2 \cap A_3|. \end{aligned}$$

$$\begin{aligned} |\overline{A_1} \cap \overline{A_2} \cap \overline{A_3}| &= 362,880 - 720 - 40,320 - 5040 \\ &\quad + 120 + 24 + 720 - 6 = 317,658. \end{aligned}$$

The Number of Onto Functions

Example 2. How many onto functions are there from set $A = \{1, 2, 3, 4, 5, 6\}$ to set $B = \{a, b, c\}$?

Sol : $f: A \rightarrow B$

$f(1) = \square$ ← $\{a, b, c\}$
 $f(2) = \square$ ←
 $\vdots$
 $f(6) = \square$ ←

不同的填法造出不同的函數
如何使 a, b, c 都出現 ?

The number of onto functions

$$\begin{aligned} &= (\text{所有函數個數}) - (a, b, c \text{ 中有一個沒被對應}) \\ &\quad + (a, b, c \text{ 中二個沒被對應}) - (a, b, c \text{ 都沒被對應}) \\ &= 3^6 - \binom{3}{1} \cdot 2^6 + \binom{3}{2} \cdot 1^6 - \binom{3}{3} \cdot 0^6 \end{aligned}$$

Thm 1. $|A| = m$, $|B| = n$

There are

$$n^m - \binom{n}{1} \cdot (n-1)^m + \binom{n}{2} \cdot (n-2)^m - \binom{n}{3} \cdot (n-3)^m + \dots \\ + (-1)^{n-1} \cdot \binom{n}{n-1} \cdot 1^m$$

onto functions $f: A \rightarrow B$.

pf :

$$A = \{a_1, a_2, \dots, a_m\}. \quad B = \{b_1, b_2, \dots, b_n\}$$
$$\begin{array}{l} f(a_1)= \square \\ f(a_2)= \square \\ \vdots \\ f(a_m)= \square \end{array} \quad \begin{array}{l} \swarrow \\ \swarrow \\ \swarrow \\ \searrow \end{array} \quad b_1, b_2, \dots, b_n$$



Example 3. How many ways are there to assign **five different jobs** to **four different employees** if *every employee is assigned at least one job*?

Sol :

Consider the assignment of jobs as a function from the set of five jobs to the set of four employees.

The number of onto functions

$$= 4^5 - \binom{4}{1} \cdot 3^5 + \binom{4}{2} \cdot 2^5 - \binom{4}{3} \cdot 1^5 = 240$$

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- **Example.** How many integers between 0 and 99,999 (inclusive) have among their digits each of 2, 5, and 8?

Sol :

$$10^5 - 3 \times 9^5 + 3 \times 8^5 - 7^5.$$

- **Example.** Determine the number of 10-combinations of the multiset

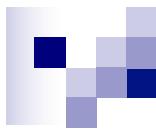
$$T = \{3 \times a, 4 \times b, 5 \times e\}.$$

- **Sol :**

$$* \text{question equals to solve } x_1 + x_2 + x_3 = 10, \text{ with } \begin{cases} 0 \leq x_1 \leq 3 \\ 0 \leq x_2 \leq 4 \\ 0 \leq x_3 \leq 5 \end{cases}$$

$$\begin{aligned} |\overline{A}_1 \cap \overline{A}_2 \cap \overline{A}_3| &= 66 - (28 + 21 + 15) + (3 + 1 + 0) - 0 \\ &= 6. \end{aligned}$$

$$\begin{aligned} &\binom{10+3-1}{2} - \left[\binom{6+3-1}{2} + \binom{5+3-1}{2} + \binom{4+3-1}{2} \right] \\ &\quad + \left[\binom{1+3-1}{2} + \binom{0+3-1}{2} + 0 \right] - 0 = 6 \end{aligned}$$



EXAMPLE 2.1.4 Find the number of solutions to $x_1 + x_2 + x_3 + x_4 = 25$, $0 \leq x_i \leq 10$. Let A_i be the solutions of $x_1 + x_2 + x_3 + x_4 = 25$ with $x_i \geq 11$. The number of solutions with $x_i \geq 0$ for all i is $\binom{25+4-1}{4-1} = \binom{25+3}{3}$. Also $|A_i| = \binom{14+3}{3}$, and $|A_i \cap A_j| = \binom{3+3}{3}$. There are no solutions with 3 or 4 of the variables larger than 10. Hence the number of solutions is

$$\binom{25+3}{3} - \binom{4}{1} \binom{14+3}{3} + \binom{4}{2} \binom{3+3}{3} = 676.$$

□



Derangements 亂序

Definition: A *derangement* is a permutation of objects that leaves ***no object in the original position.***

Example 5: The permutation of 21453 is a derangement of 12345 because no number is left in its original position. But 21543 is not a derangement of 12345, because 4 is in its original position.



Def.

Let D_n be the number of derangements of n objects.

$$\begin{aligned} D_4 = & \text{(所有4個元素的permutation數)} \\ & - \text{(4個元素有一個在原位置的permutation數)} \\ & + \text{(4元素中有二個在原位置的個數)} \\ & - \text{(4個元素中有三個在原位置的個數)} \\ & + \text{(4元素都在原位置的個數)} \end{aligned}$$

$$= 4! - \binom{4}{1} \cdot 3! + \binom{4}{2} \cdot 2! - \binom{4}{3} \cdot 1! + \binom{4}{4} \cdot 0!$$

Theorem 2. (亂序公式)

$$\begin{aligned} D_n &= n! - \binom{n}{1} \cdot (n-1)! + \binom{n}{2} \cdot (n-2)! - \dots + (-1)^n \binom{n}{n} \cdot 0! \\ &= n! \left(1 - \frac{n!}{1!(n-1)!} \cdot \frac{(n-1)!}{n!} + \frac{n!}{2!(n-2)!} \cdot \frac{(n-2)!}{n!} - \dots \right. \\ &\quad \left. + (-1)^n \cdot \frac{n!}{n!0!} \cdot \frac{0!}{n!} \right) \\ &= n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \dots + (-1)^n \cdot \frac{1}{n!} \right) \\ &= n! \sum_{k=0}^n (-1)^k \frac{1}{k!} = \sum_{k=0}^n (-1)^k \frac{1}{k!} \end{aligned}$$

Derangements

The Hatcheck Problem: A new employee checks the hats of n people at restaurant, forgetting to put claim check numbers on the hats. When customers return for their hats, the checker gives them back hats chosen at random from the remaining hats. What is the probability that no one receives the correct hat.

Solution: The answer is the number of ways the hats can be arranged so that there is no hat in its original position divided by $n!$, the number of permutations of n hats.

Remark: It can be shown that the probability of a derangement approaches $e^{-1} \approx 0.368$ as n grows without bound.

$$\frac{D_n}{n!} = \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + (-1)^n \frac{1}{n!} \right]$$
$$\left(= \frac{1}{n!} n! \sum_{k=0}^n (-1)^k \frac{1}{k!} = \sum_{k=0}^n (-1)^k \frac{1}{k!} \right)$$

TABLE 1 The Probability of a Derangement.

n	2	3	4	5	6	7
$D_n / n!$	0.50000	0.33333	0.37500	0.36667	0.36806	0.36786

Exercise.

How many ways can the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 be arranged so that **even** digit is in its original position?

Sol :

$$10! - \binom{5}{1} \cdot 9! + \binom{5}{2} \cdot 8! - \binom{5}{3} \cdot 7! + \binom{5}{4} \cdot 6! - \binom{5}{5} \cdot 5!$$

EXAMPLE 2.2.1 How many permutations of $[n]$ contain no instance of i followed by $i + 1$?

By a similar use of the inclusion-exclusion formula, it turns out that this is

$$Q_n = n! \sum_{k=0}^{n-1} (-1)^k \frac{1}{k!} + (n-1)! \sum_{k=1}^{n-1} (-1)^{k-1} \frac{1}{(k-1)!}.$$

Note that the limits on the two sums are not identical.

□